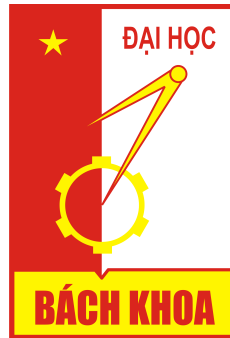


HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF ELECTRICAL AND ELECTRONIC ENGINEERING



GRADUATION THESIS

SIMULATION-BASED 5G RADIO MEASUREMENT DATA GENERATION AND LOG-BASED HANDOVER DELAY ANALYSIS

Student: Hoang Gia Duc
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Class: ET-E4 01 K67

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Hanoi, June 28, 2026

ĐÁNH GIÁ QUYỂN ĐỒ ÁN TỐT NGHIỆP

(Dùng cho giảng viên hướng dẫn)

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4	Có kết quả mô phỏng/thực nghiệm và trình bày rõ ràng kết quả đạt được	1	2	3	4 5
Có khả năng phân tích và đánh giá kết quả (15)					
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6	Kết quả được trình bày một cách logic và dễ hiểu, tất cả kết quả đều được phân tích và đánh giá thỏa đáng	1	2	3	4 5
7	Trong phần kết luận, tác giả chỉ rõ sự khác biệt (nếu có) giữa kết quả đạt được và mục tiêu ban đầu đề ra đồng thời cung cấp lập luận để đề xuất hướng giải quyết có thể thực hiện trong tương lai	1	2	3	4 5
Kỹ năng viết quyển đồ án (10)					
8	Đồ án trình bày đúng mẫu quy định với cấu trúc các chương logic và đẹp mắt (bảng biểu, hình ảnh rõ ràng, có tiêu đề, được đánh số thứ tự và được giải thích hay đề cập đến; căn lề thống nhất, có dấu cách sau dấu chấm, dấu phẩy v.v.), có mở đầu chương và kết luận chương, có liệt kê tài liệu tham khảo và có trích dẫn đúng quy định	1	2	3	4 5
9	Kỹ năng viết xuất sắc (cấu trúc câu chuẩn, văn phong khoa học, lập luận logic và có cơ sở, từ vựng sử dụng phù hợp v.v.)	1	2	3	4 5
Thành tựu nghiên cứu khoa học (5) (chọn 1 trong 3 trường hợp)					
10a	Có bài báo khoa học được đăng hoặc chấp nhận đăng/Đạt giải SVNCKH giải 3 cấp Viện trở lên/Có giải thưởng khoa học (quốc tế hoặc trong nước) từ giải 3 trở lên/Có đăng ký bằng phát minh, sáng chế	5			
10b	Được báo cáo tại hội đồng cấp Viện trong hội nghị SVNCKH nhưng không đạt giải từ giải 3 trở lên/Đạt giải khuyến khích trong các kỳ thi quốc gia và quốc tế khác về chuyên ngành (VD: TI contest)	2			
10c	Không có thành tích về nghiên cứu khoa học	0			
Điểm tổng					/50
Điểm tổng quy đổi về thang 10					

***Nhận xét khác** (về thái độ và tinh thần làm việc của sinh viên)*

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Người nhận xét
(Ký và ghi rõ họ tên)

ĐÁNH GIÁ QUYỂN ĐỒ ÁN TỐT NGHIỆP

(Dùng cho cán bộ phản biện)

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Điểm tổng quy đổi về thang 10						

Nhận xét khác của cán bộ phản biện

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Người nhận xét
(Ký và ghi rõ họ tên)

PREFACE

As fifth-generation (5G) mobile networks continue to expand, mobility management has become an important aspect of maintaining service continuity and network performance. Among mobility procedures, handover plays a key role because user equipment may need to move between serving cells while radio conditions and network conditions change over time. This topic is important not only from the viewpoint of radio coverage, but also from the viewpoint of signaling procedures and timing behavior inside the network.

However, collecting detailed 5G mobility data in real environments is difficult because measurement campaigns require suitable equipment, stable test conditions, and repeated experiments. In addition, handover behavior cannot be fully understood from radio measurements alone, since a change in the best serving cell does not necessarily mean that the protocol-level handover procedure has been completed. These difficulties motivated the development of a controlled study environment in which radio measurements and handover-delay behavior can be inspected more systematically.

For that reason, this graduation thesis was developed to support the study of 5G mobility through simulation-based radio-data generation and log-based handover-delay analysis. The work focuses on generating structured radio-measurement data and examining handover delay from timer-event logs under different timer configurations and network-condition settings.

Through this work, radio-level serving-cell decisions and protocol-level handover completion can be considered separately. This separation provides a clearer basis for observing handover behavior and for evaluating how timer configurations and network conditions may influence completion time, timeout cases, and failure behavior.

I would like to express my deepest gratitude to **Dr. Phung Thi Kieu Ha** for her guidance, feedback, and support throughout this graduation thesis. I also thank the lecturers of the School of Electrical and Electronic Engineering, Hanoi University of Science and Technology, for providing the technical and academic foundation necessary for this work. Due to limitations in time, experimental resources, and experience, some limitations may remain. I sincerely welcome constructive comments and suggestions that can improve both the thesis and the research platform.

DECLARATION OF AUTHORSHIP

I am Hoang Gia Duc, student ID 20224339, a student of the Advanced Program in Electronics and Telecommunication Engineering, cohort K67. I declare that the thesis entitled "**Simulation-Based 5G Radio Measurement Data Generation and Log-Based Handover Delay Analysis**" was carried out by myself under the supervision of **Dr. Phung Thi Kieu Ha**. All data, results, and evaluations presented in this thesis are reported honestly and reflect the actual outcomes obtained during the research process. All referenced documents, source code, standards, and technical materials have been properly cited and listed in accordance with academic and intellectual property regulations. I take full responsibility for the content of this thesis.

Hanoi, June 28, 2026

Author

A handwritten signature in black ink, appearing to be 'HGD' with a stylized flourish at the end.

Hoang Gia Duc

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ABBREVIATIONS

3GPP	Third Generation Partnership Project
5G	Fifth Generation
5GC	5G Core
A2G	Air-to-Ground
AMF	Access and Mobility Management Function
BS	Base Station
CHO	Conditional Handover
CSV	Comma-Separated Values
DL	Downlink
eMBB	enhanced Mobile Broadband
FSM	Finite-State Machine
gNB	next-generation NodeB
GTP-U	GPRS Tunnelling Protocol for the User Plane
HO	Handover
HOM	Handover Margin
ISD	Inter-Site Distance
LOS	Line-of-Sight
mMTC	massive Machine-Type Communications
N2	Control-plane interface between a gNB and the AMF
N3	User-plane interface between a gNB and the UPF
NAS	Non-Access Stratum
NGAP	NG Application Protocol
NLOS	Non-Line-of-Sight
NTN	Non-Terrestrial Network
PDU	Protocol Data Unit
PLMN	Public Land Mobile Network
RAN	Radio Access Network
RB	Resource Block
RLF	Radio Link Failure
RRC	Radio Resource Control
RSRP	Reference Signal Received Power
SCTP	Stream Control Transmission Protocol
SINR	Signal-to-Interference-plus-Noise Ratio
SMF	Session Management Function
TN	Terrestrial Network

UAV	Unmanned Aerial Vehicle
UE	User Equipment
UL	Uplink
UPF	User Plane Function
URLLC	Ultra-Reliable Low-Latency Communications
Xn	Logical interface between gNBs

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ABSTRACT

This thesis investigates mobility behavior in fifth-generation (5G) networks through radio-measurement data generation and handover-delay analysis. Since real-world 5G measurement campaigns require specialized equipment, repeated field tests, and stable experimental conditions, collecting detailed mobility datasets remains challenging. In addition, handover performance cannot be evaluated only from serving-cell changes, because a radio-level decision does not necessarily indicate the completion of the protocol-level handover procedure.

To address these issues, this thesis develops a simulation-based radio-data generation workflow and a log-based handover-delay analysis workflow. The first part generates structured 5G radio-measurement data during UE movement, including mobility information, serving-cell changes, radio-channel characteristics, and handover-related indicators. The second part analyzes handover delay using timer-event logs under different timer configurations and network-condition settings.

The proposed workflow separates radio-level serving-cell decisions from protocol-level handover completion. This separation enables clearer observation of handover behavior and provides a basis for evaluating completion time, timeout cases, failure behavior, and recovery margin. The results show that timer configuration and network conditions have an important influence on handover performance, especially in cases where signaling delay or packet loss affects the completion process.

Overall, this thesis provides a practical platform for generating 5G mobility data and analyzing handover delay beyond simple success-or-failure outcomes. The workflow can support further experiments on mobility optimization, timer configuration, and data-driven handover analysis.

CHAPTER 1. INTRODUCTION

1.1 Research Context

Mobility management is an essential function in 5G networks because user equipment (UE) is expected to maintain service continuity while moving through an environment with continuously changing radio conditions. In a mobile scenario, the radio link between a UE and surrounding gNBs is not fixed. As the UE changes its position or height, the distance to each gNB, the line-of-sight (LOS) condition, the path loss, the shadow fading component, and the interference level can all vary over time. These variations directly affect radio measurements such as reference signal received power (RSRP), received signal strength, and signal-to-interference-plus-noise ratio (SINR) [1]. Since these measurements are commonly used to evaluate serving-cell quality, they provide an important basis for cell selection, cell reselection, and handover-related decisions.

In dense 5G deployments, this problem becomes more significant because the UE may observe several candidate cells with comparable signal quality. A small change in position may improve the received signal from one gNB while reducing the quality of another. Similarly, a temporary obstruction or a rapid change in interference may cause the best serving-cell candidate to change within a short time interval. If mobility decisions are made too late, the UE may experience degraded signal quality before the handover is completed. If they are made too aggressively, unnecessary handovers may occur, increasing signaling overhead and possibly causing instability. Therefore, studying mobility behavior requires not only observing the final serving cell, but also understanding how the underlying radio measurements evolve before, during, and after a potential handover event.

At the same time, handover is not only a radio-measurement problem. Radio measurements may indicate that a target cell is more suitable, but the actual handover still requires a protocol procedure involving the source gNB, target gNB, UE, and 5G core network (5GC) [2]. After the target cell is selected, signaling messages must be exchanged, resources must be prepared, and timers must be monitored until the procedure either completes successfully or fails. The final handover delay can therefore be affected by transport latency, packet loss, jitter, and protocol timer configuration. For this reason, a more complete study of mobility should consider both radio-level measurements and protocol-level handover execution. The radio layer explains why a serving-cell change may be needed, while the protocol layer explains how long the handover procedure takes and whether it can complete under the given network conditions.

1.2 Problem Statement

Collecting real-world mobile-network data requires a dedicated measurement setup, repeated field tests, and considerable time and effort. Existing measurement datasets illustrate both the value and the cost of this approach. For example, Meixner et al. provide a 4G/LTE channel-quality trace dataset containing hundreds of LTE traces with reference-signal indicators such as RSRP and RSRQ, collected across different environments, speeds, network providers, and countries [3]. Such traces are useful because they capture radio behavior under practical deployment conditions. However, measurement traces are observational: they record what happened during specific field campaigns, but they do not allow the researcher to freely change one radio or mobility factor while keeping the remaining conditions under direct control. As a result, real-world measurement data is valuable, but it is not always convenient for isolating the influence of individual radio factors.

Another limitation is that the variables recorded in a measurement dataset do not always match the variables required by a particular mobility study. A trace may include useful channel-quality or reference-signal measurements, but may not include all intermediate quantities used in this thesis, such as path loss, shadow fading, per-cell RSRP, SINR, serving-cell transitions, or handover-related labels. In addition, radio measurements alone do not show the signaling events and timer states needed to analyze protocol-level handover delay. For example, they may show that the serving cell changed, but not when the handover request was sent, when the UE completed access to the target gNB, whether a timer expired, or which signaling phase caused most of the delay. This makes it difficult to compare parameter settings systematically or analyze how individual radio and protocol factors contribute to mobility behavior. A controlled simulation and emulation workflow is therefore useful because it allows radio parameters, mobility settings, protocol events, and output variables to be defined explicitly and recorded in a structured format.

The main problem is therefore the separation between radio-level serving-cell changes and protocol-level handover completion. A radio trace may indicate that another gNB becomes a better serving candidate based on RSRP or SINR, but this does not mean that the handover procedure has already completed. In a real or emulated network procedure, the UE, source gNB, target gNB, and core network still need to complete the required signaling steps before the handover is finalized. If this distinction is not made, the analysis may incorrectly treat a radio decision as a completed handover. This motivates a workflow that can generate structured radio data and analyze handover delay from protocol logs, while keeping the radio-level and protocol-level views separate.

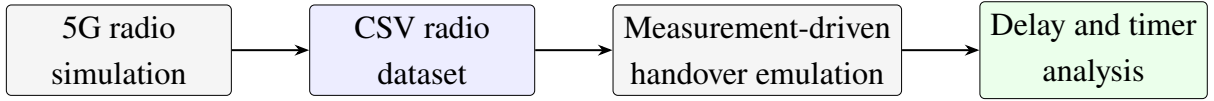


Figure 1.1 Overall workflow of radio-data generation and handover-delay analysis

1.3 Objectives

The objective of this thesis is to build and evaluate a configurable workflow for studying 5G radio measurements and handover delay. The work is designed around two connected parts. The first part focuses on generating radio-measurement data in a controlled simulation environment. The second part focuses on using measurement-driven handover experiments and logs to analyze the delay of the handover procedure. By combining these two parts, the thesis aims to provide a clearer view of how radio conditions motivate mobility decisions and how protocol execution determines the final handover completion behavior.

From a functional perspective, the workflow should be able to simulate UE movement, calculate relevant radio parameters, select or record serving-cell behavior, and export the generated measurements into a structured CSV format. It should also support handover-delay experiments in which timer events and network-condition effects can be inspected. These functional requirements are important because the output of the system must be directly usable for later analysis, instead of only producing visual observations or final summary values.

From a non-functional perspective, the workflow should be configurable, repeatable at the experiment-design level, and suitable for systematic post-processing. Repeatability in this thesis means that the experiment settings, execution steps, and output fields are defined consistently across runs, even though the generated UE trajectory may differ when random mobility is enabled. Configurability is needed because different UE heights, movement settings, radio conditions, and network-delay settings may be examined. The output format should also be clear enough to support post-processing, comparison, and visualization without requiring manual interpretation of unstructured logs.

The specific objectives are:

- generate structured radio-measurement data from a controllable 5G propagation simulation;
- record key mobility and radio parameters, including UE position, serving cell, RSRP, SINR, and handover indicators;
- use measurement-driven handover experiments to observe protocol-level handover

behavior;

- analyze how timer behavior and network conditions affect handover completion time, timeout cases, and failure behavior.

1.4 Scope

The thesis focuses on simulation-based radio-data generation and log-based handover-delay analysis. The radio-data generation part is limited to the implemented 5G propagation and mobility simulation workflow. It considers the parameters that are available in the simulator, such as UE position, gNB position, propagation-related values, serving-cell information, RSRP, SINR, and handover-related indicators. The purpose of this part is not to reproduce every detail of a commercial 5G network, but to generate structured data from explicitly defined simulation settings so that mobility behavior can be inspected under selected scenarios.

The handover-delay analysis part is limited to the implemented measurement-driven handover-emulation workflow and the available experiment logs. The analysis focuses on completion time, timeout behavior, failure cases, and the influence of network conditions such as latency, packet loss, and jitter. It does not attempt to model every internal mechanism of a full 5G core deployment or every vendor-specific handover implementation. Instead, it uses the available timer and event logs to study how protocol-level execution contributes to the observed handover delay.

This thesis does not attempt to perform a full real-world 5G measurement campaign. Such a campaign would require specialized equipment, controlled field conditions, repeated physical measurements, and access to operational network configurations. The thesis also does not aim to design a complete commercial handover optimization algorithm. Although the generated data and delay analysis can support future optimization work, the main focus here is on building the workflow, generating structured data, and analyzing the resulting handover delay characteristics.

1.5 Main Contributions

The main contributions of this thesis are:

- a configurable simulation pipeline for generating structured 5G radio-measurement traces;
- a measurement-driven workflow that separates radio-level serving-cell changes from protocol-level handover execution;
- a log-based analysis of handover delay, timer behavior, timeout cases, and failure behavior under selected network conditions.

The first contribution is the construction of a configurable radio-data generation pipeline. The pipeline uses explicitly defined simulation settings to generate mobility and radio-measurement traces, including UE position, serving-cell information, RSRP, SINR, and handover-related indicators. This provides structured data that can be inspected and processed without relying only on manually collected field traces.

The second contribution is the separation between radio-level and protocol-level mobility views. The generated radio data can indicate when a serving-cell change is suggested by radio measurements, while the handover-emulation workflow is used to observe whether and how the corresponding protocol-level handover procedure completes. This separation reduces ambiguity between a radio decision and a completed handover.

The third contribution is the analysis of handover delay and timer behavior using available experiment logs. The thesis examines completion time, timeout cases, failure behavior, and the influence of selected network impairments such as latency, packet loss, and jitter. This provides a more detailed view of handover performance than a simple success-or-failure summary.

1.6 Chapter Conclusion

This chapter has established the motivation and boundaries of the thesis by clarifying the mobility problem, the need to separate radio-level observations from protocol-level handover behavior, and the intended contributions of the work. The remainder of this thesis is organized as follows:

Chapter 2. Cellular Network and Mobility Management

Chapter 3. Investigation of Mobility Management in Aerial-to-Ground Communications

Chapter 4. Experimental Results

Chapter 5. Conclusion

CHAPTER 2. CELLULAR NETWORK AND MOBILITY MANAGEMENT

2.1 Overview of Cellular Mobile Networks and Air-to-Ground Communication

Cellular mobile networks form the basic architecture of modern mobile communication systems. In this architecture, a wide geographical service area is divided into smaller radio coverage regions called cells. Each cell is served by one or more base stations (BSs), and the UE accesses the mobile network through the wireless link provided by the serving BS [4]. This cell-based structure allows the network to provide wide-area coverage while reusing limited radio spectrum across different locations.

The main elements of a cellular mobile network include the UE, the radio access network (RAN), the BS, and the core network. The UE represents the terminal device used by the subscriber, such as a smartphone, an Internet of Things (IoT) device, or an aerial terminal. The RAN provides the radio interface between the UE and the network, while the BS controls radio transmission, reception, and radio-resource management inside its coverage area. The core network performs higher-level functions such as session management, mobility support, authentication, and connectivity to external data networks.

A key characteristic of cellular systems is mobility support. Since a UE may move within a cell or from one cell to another, the network must continuously maintain service continuity and decide which BS should serve the UE. When the UE moves toward a neighboring cell and the serving condition changes, a handover (HO) procedure may be performed to transfer the connection from the current serving cell to a target cell. Therefore, the cellular concept is closely related to coverage planning, neighboring-cell relationships, radio-resource management, and mobility management.

Based on this cellular principle, mobile networks have evolved from early voice-oriented systems to broadband and intelligent communication systems. In 5G, the cellular architecture is extended by the New Radio (NR) air interface, gNB-based radio access, dense deployment, massive multiple-input multiple-output (MIMO), beamforming, network slicing, and 5G Core (5GC) connectivity [2]. Looking beyond 5G, 6G is expected to further enhance cellular networks by integrating artificial intelligence, sensing, ultra-dense connectivity, and new spectrum technologies to support more advanced communication scenarios. This 6G perspective is mentioned only as background motivation, because the experimental work in this thesis remains focused on 5G-oriented radio measurements and handover behavior.

The 5G mobile network is designed to support three main service objectives: En-

hanced Mobile Broadband (eMBB) for high data rate services, Ultra-Reliable Low Latency Communications (URLLC) for delay-sensitive and reliable services, and Massive Machine-Type Communications (mMTC) for connecting a large number of devices [4]. In a basic 5G system, the UE is the terminal device that accesses the mobile network. A UE can be a smartphone, an Internet of Things (IoT) device, or in an aerial communication scenario, an unmanned aerial vehicle (UAV) acting as an aerial UE.

The UE connects to the RAN through a BS, referred to in 5G as a gNB. The gNB is responsible for providing the wireless radio interface, transmitting and receiving radio signals, managing radio resources, and maintaining the radio connection with the UE. It also forwards user data and control information between the UE and the 5GC network. Therefore, the gNB is the main network element that directly determines how the UE accesses the 5G system over the air interface.

In a conventional terrestrial scenario, the UE is usually located near the ground and moves inside the coverage area of surrounding gNBs. The radio link quality is mainly affected by the horizontal distance between the UE and the serving gNB, obstacles such as buildings or terrain, shadow fading, and interference from neighboring cells.

Air-to-ground (A2G) communication extends this model by considering a UAV or aerial UE communicating with ground-based gNBs. This corresponds to the cellular-connected UAV case, where the UAV acts as a flying mobile terminal within a cellular network [5, 6]. Unlike a ground UE, an aerial UE operates at a certain altitude and can have LOS or near-LOS paths to several BSs. This difference changes the radio propagation condition because the signal quality is affected not only by horizontal distance, but also by UE altitude, three-dimensional link distance, LOS condition, and interference from neighboring gNBs [6].

As the UAV moves, these link conditions can change continuously. A higher altitude may improve visibility to the serving gNB, but it may also expose the UAV to stronger interference from multiple neighboring gNBs; this LOS-dominant channel and aerial-terrestrial interference are highlighted as key issues for UAV communications in 5G and beyond [6]. For this reason, received signal levels and link conditions may vary rapidly in A2G scenarios. These variations make A2G communication an important case for studying radio measurement behavior and mobility-related problems in 5G networks.

Figure 2.1 illustrates the A2G geometry considered in this thesis. It connects the main physical quantities used later in the radio-measurement model: UE height, horizontal distance, three-dimensional link distance, the serving link, and interference from neighboring gNBs.

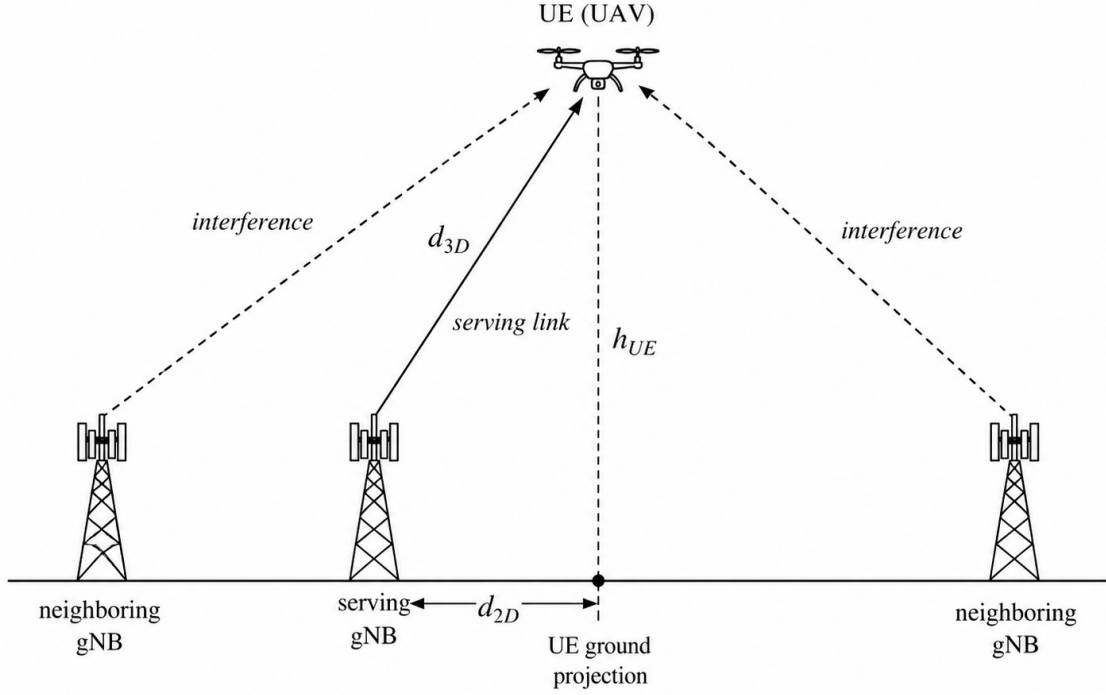


Figure 2.1 A2G geometry used to interpret UE height, serving-link distance, and neighboring-cell interference

The network layout determines the relative positions of serving and neighboring gNBs. It directly influences propagation distance, received signal strength, interference, and mobility-related events. This thesis uses a hexagonal cellular layout as the theoretical basis for representing the network geometry.

The hexagonal cellular model is a conceptual approximation used to represent the radio coverage of a BS. In an ideal isotropic environment, equal received-power contours around a BS would be approximately circular. Actual cell boundaries, however, are irregular because they are shaped by terrain, buildings, antenna patterns, propagation loss, and interference. Therefore, a hexagonal cell should not be interpreted as the exact physical coverage boundary of a gNB, but as a simplified geometric abstraction for network planning and analysis [7].

Circular cells cannot cover a large two-dimensional area without leaving gaps or creating overlaps. Regular hexagons, in contrast, tessellate the plane exactly and approximate circular coverage more closely than the other regular polygons that can tessellate the plane [7]. The hexagonal representation consequently provides both a simple approximation of radio coverage and a consistent structure for placing neighboring gNBs. Each

cell has six immediate neighbors, which facilitates the analysis of inter-cell interference, cell association, and HO behavior.

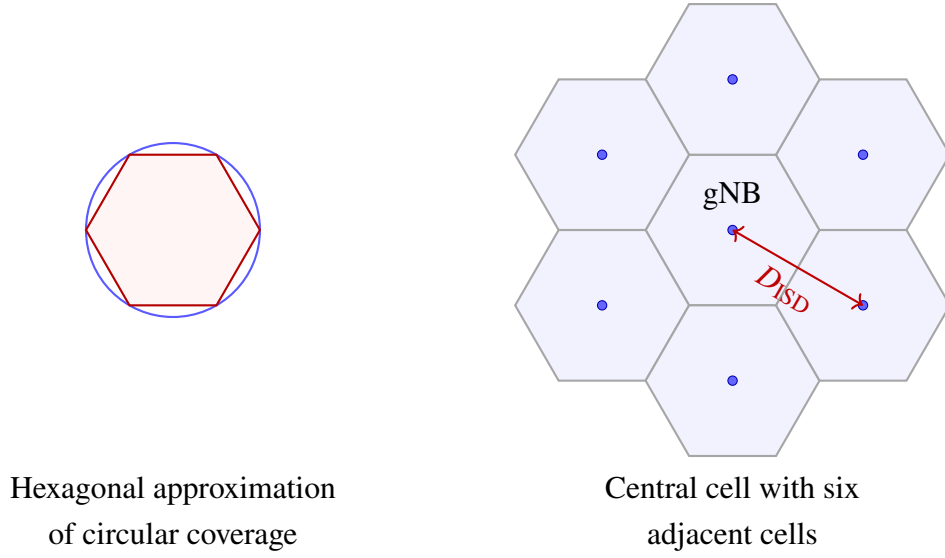


Figure 2.2 Circular coverage approximation and seven-cell hexagonal layout

The distance between the centers of two adjacent cells is commonly referred to as the inter-site distance (ISD). For a regular hexagonal cell with radius R , measured from the center to a vertex, the ISD is

$$D_{\text{ISD}} = \sqrt{3}R. \quad (2.1)$$

The cell radius and ISD determine the spatial density of gNB deployment. A smaller ISD produces a denser network, which may improve coverage but also increases the possibility of inter-cell interference. Conversely, a larger ISD reduces the number of gNBs in a given area but may result in weaker received power near cell boundaries. Therefore, the hexagonal model is useful for studying coverage, neighboring-cell interference, cell association, and HO behavior.

2.2 Radio Propagation in 5G Networks and Measurement Metrics

Radio propagation describes how the transmitted electromagnetic signal changes before it reaches the receiver. In a 5G cellular network, the received signal is affected by distance, carrier frequency, antenna gains, LOS or NLOS condition, shadowing objects, interference from other cells, and receiver noise. 3GPP TR 38.901 provides the commonly used channel-modelling framework for frequencies from 0.5 GHz to 100 GHz, including path loss, LOS probability, penetration loss, shadow fading, and fast-fading components for scenarios such as Urban Macro (UMa) and Urban Micro (UMi) [8]. This thesis uses these concepts at a system-simulation level to generate interpretable

radio measurements rather than to reproduce a full link-level channel realization.

For a UE and a gNB separated by horizontal distance d_{2D} , the three-dimensional link distance is

$$d_{3D} = \sqrt{d_{2D}^2 + (h_{\text{UE}} - h_{\text{gNB}})^2}, \quad (2.2)$$

where h_{UE} and h_{gNB} are the UE and gNB antenna heights, respectively. This distance is especially important in A2G scenarios because a change in UE height changes both the link distance and the elevation geometry. A higher aerial UE may have a higher LOS probability to several gNBs, but the same visibility can also increase received interference from neighboring cells.

2.2.1 Large-Scale Propagation Model for Mobile Wireless Communication

The radio simulator uses a large-scale propagation abstraction rather than a full small-scale fading channel. This abstraction is appropriate for generating mobility traces because the quantities of interest are serving-cell strength, neighboring-cell strength, and SINR trends along the UE trajectory. The theoretical chain used in the simulator is

$$\{d_{2D}, d_{3D}, h_{\text{UE}}\} \rightarrow P_{\text{LOS}} \rightarrow S_{\text{LOS/NLOS}} \rightarrow PL \rightarrow X_{\sigma} \rightarrow \{P_{\text{rx}}, \text{RSRP}, \text{SINR}\}. \quad (2.3)$$

In this chain, geometry determines path length and LOS probability. The sampled LOS/NLOS state selects the path-loss formula. Shadow fading adds a spatially varying large-scale random term. Finally, the link budget and interference model convert these losses into received power, RSRP, and SINR. Table 2.1 links each theoretical component to the corresponding dataset output used in the thesis.

Table 2.1 Theoretical propagation components used by the radio simulator [8, 9, 1]

Component	Theoretical role	Dataset output
Link geometry	Defines UE and gNB positions, UE height, movement state, and the internally computed 2D/3D link distance.	UE position, UE height, direction, speed, connected BS; internal 2D/3D distance
LOS/NLOS condition	Represents the probability and realization of direct visibility between the UE and each gNB.	LOS probability, LOS state
Path loss	Models median large-scale attenuation due to distance, carrier frequency, height, and LOS/NLOS condition.	Path loss
Shadow fading	Models slow large-scale random variation around the median path loss caused by environmental obstruction.	Shadow-fading value
Link budget and RSRP	Combines transmit power, antenna gains, path loss, and shadow fading to obtain received power and reference-signal strength.	Received power, serving-cell RSRP, neighboring-cell RSRP
Interference and noise	Combines neighboring-cell received powers with thermal noise to evaluate the serving-link quality.	SINR

2.2.2 *LOS Probability and LOS/NLOS State*

In radio propagation, a line-of-sight (LOS) link refers to a condition in which the direct propagation path between the transmitter and receiver is not blocked by major obstacles. A non-line-of-sight (NLOS) link refers to a condition in which the direct path is obstructed, so the received signal is mainly affected by reflection, diffraction, scattering, and additional attenuation. This distinction is important because LOS and NLOS links generally have different path-loss behavior, and the simulator selects the corresponding path-loss expression according to the realized link state.

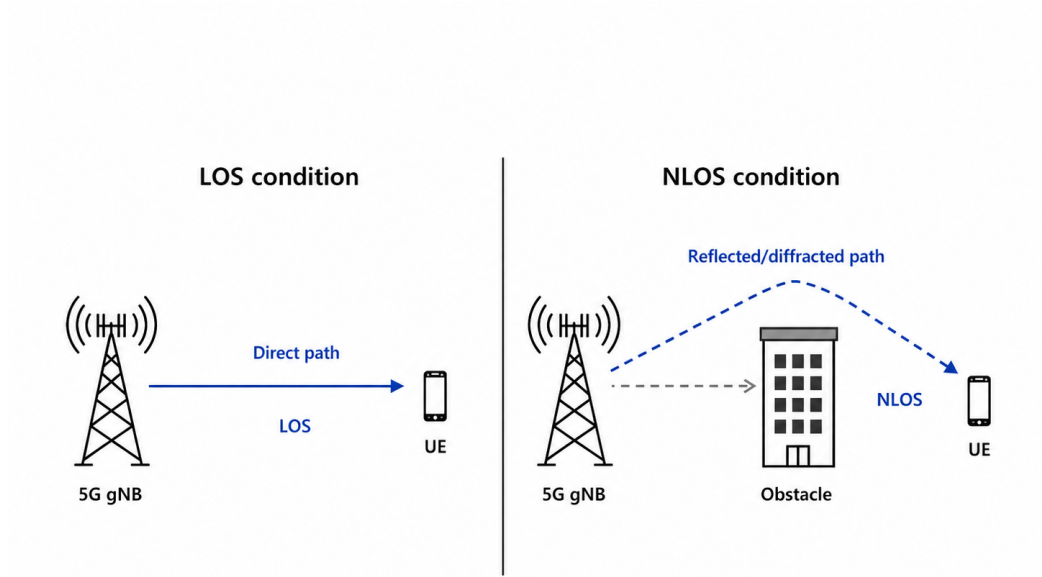


Figure 2.3 Illustration of LOS and NLOS propagation conditions between a 5G gNB and a UE

The simulator first calculates a LOS probability for each UE–gNB link. The model follows the UMa LOS probability in 3GPP TR 38.901 for terrestrial and low-height UEs and the UMa-AV extension in 3GPP TR 36.777 for aerial UEs [8, 9]. The NLOS probability is then

$$P_{\text{NLOS}} = 1 - P_{\text{LOS}}. \quad (2.4)$$

All distances in the following LOS-probability expressions are in metres.

The implemented LOS-probability calculation can be followed from the UE height range and then from the horizontal distance d_{2D} . For terrestrial and low-height UEs, $1.5 \leq h_{\text{UE}} \leq 22.5$ m, the simulator uses the UMa expression from 3GPP TR 38.901 [8]. If the UE is very close to the gNB in the horizontal plane, $d_{2D} \leq 18$ m, the link is treated as LOS with

$$P_{\text{LOS}} = 1. \quad (2.5)$$

When $d_{2D} > 18$ m, the LOS probability decreases with distance but includes a height correction for UEs above 13 m. In the following expressions, $\text{clip}(x, 0, 1)$ denotes limiting x to the interval $[0, 1]$:

$$P_{\text{LOS}} = \text{clip}(P_{\text{base}} G_h, 0, 1), \quad (2.6)$$

$$P_{\text{base}} = \frac{18}{d_{2D}} + \left(1 - \frac{18}{d_{2D}}\right) e^{-d_{2D}/63}, \quad (2.7)$$

$$G_h = 1 + \frac{5}{4} C(h_{\text{UE}}) \left(\frac{d_{2D}}{100}\right)^3 e^{-d_{2D}/150}. \quad (2.8)$$

The height correction is inactive for ordinary ground-level UEs and becomes active only when $h_{\text{UE}} > 13$ m:

$$C(h_{\text{UE}}) = \begin{cases} 0, & h_{\text{UE}} \leq 13, \\ \left(\frac{h_{\text{UE}} - 13}{10}\right)^{1.5}, & 13 < h_{\text{UE}} \leq 22.5. \end{cases} \quad (2.9)$$

For aerial UEs in the range $22.5 < h_{\text{UE}} \leq 100$ m, the simulator switches to the UMa-AV LOS-probability model from 3GPP TR 36.777 [9]. It first calculates two height-dependent parameters:

$$d_1 = \max(460 \log_{10}(h_{\text{UE}}) - 700, 18), \quad (2.10)$$

$$p_1 = 4300 \log_{10}(h_{\text{UE}}) - 3800. \quad (2.11)$$

The value d_1 acts as the first horizontal-distance threshold. If $d_{2D} \leq d_1$, the link is treated as LOS:

$$P_{\text{LOS}} = 1. \quad (2.12)$$

If $d_{2D} > d_1$, the LOS probability is reduced according to

$$P_{\text{LOS}} = \text{clip}\left(\frac{d_1}{d_{2D}} + e^{-d_{2D}/p_1} \left(1 - \frac{d_1}{d_{2D}}\right), 0, 1\right). \quad (2.13)$$

Finally, for high aerial UEs, $100 < h_{\text{UE}} \leq 300$ m, the implemented model sets

$$P_{\text{LOS}} = 1. \quad (2.14)$$

After P_{LOS} is calculated, the simulator converts this probability into a concrete LOS/NLOS state by Bernoulli sampling. For each UE-gNB link, a uniform random variable $u \sim \mathcal{U}(0, 1)$ is drawn and the link state S is assigned as

$$S = \begin{cases} \text{LOS}, & u \leq P_{\text{LOS}}, \\ \text{NLOS}, & u > P_{\text{LOS}}. \end{cases} \quad (2.15)$$

Thus, a larger P_{LOS} gives a higher chance that the link is realized as LOS, while $P_{\text{LOS}} = 1$ always gives a LOS link and $P_{\text{LOS}} = 0$ always gives a NLOS link. The implemented simulator keeps the sampled LOS/NLOS state for each UE–gNB link and reuses it in the subsequent path-loss and shadow-fading calculations [9]. This avoids artificial step-to-step switching that would occur if the LOS/NLOS state were redrawn independently at every simulation step. This is a modelling simplification for generating a stable trace, not a statement that the physical LOS/NLOS condition in a real environment can never change.

2.2.3 Path-Loss Model

Path loss represents the large-scale attenuation between the transmitter and receiver. It increases with distance and depends on the propagation environment, carrier frequency, UE height, and LOS/NLOS state. The simulator follows a height-driven calculation order: first it prepares effective distances, then it selects the UMa or UMa-AV formula according to UE height and sampled LOS/NLOS state. The low-height UMa formulas are based on 3GPP TR 38.901, and the aerial UMa-AV formulas are based on 3GPP TR 36.777 [8, 9].

Before evaluating path loss, the horizontal distance is lower-bounded by 10 m and the effective 3D distance is recomputed:

$$d_{2D,\text{eff}} = \max(d_{2D}, 10), \quad (2.16)$$

$$d_{3D,\text{eff}} = \sqrt{d_{2D,\text{eff}}^2 + (h_{\text{gNB}} - h_{\text{UE}})^2}. \quad (2.17)$$

For readability, the following path-loss equations use d_{2D} and d_{3D} to denote these effective distances after preprocessing.

The UMa LOS formula also uses the effective breakpoint distance

$$d'_{\text{BP}} = \frac{4(h_{\text{gNB}} - 1)(h_{\text{UE}} - 1)f_c}{c}, \quad (2.18)$$

where f_c is in Hz and c is the speed of light. In the remaining path-loss formulas, $f_{c,\text{GHz}}$ denotes the carrier frequency in GHz.

For $1.5 \leq h_{\text{UE}} \leq 22.5$ m, the simulator uses the UMa path-loss model. If the sampled link state is LOS and $d_{2D} \leq d'_{\text{BP}}$, the LOS path loss is

$$PL_{\text{LOS}} = 28 + 22\log_{10}(d_{3D}) + 20\log_{10}(f_{c,\text{GHz}}). \quad (2.19)$$

If the same low-height LOS link is beyond the breakpoint, $d_{2D} > d'_{\text{BP}}$, the slope changes to

$$PL_{\text{LOS}} = 28 + 40\log_{10}(d_{3D}) + 20\log_{10}(f_{c,\text{GHz}}) - 9\log_{10}((d'_{\text{BP}})^2 + (h_{\text{gNB}} - h_{\text{UE}})^2). \quad (2.20)$$

If the sampled low-height link state is NLOS, the simulator first computes

$$PL'_{\text{NLOS}} = 13.54 + 39.08\log_{10}(d_{3D}) + 20\log_{10}(f_{c,\text{GHz}}) - 0.6(h_{\text{UE}} - 1.5), \quad (2.21)$$

and then applies the 3GPP max operation:

$$PL_{\text{NLOS}} = \max(PL_{\text{LOS}}, PL'_{\text{NLOS}}). \quad (2.22)$$

For aerial UEs, $22.5 < h_{\text{UE}} \leq 300$ m, the LOS path-loss expression becomes the UMa-AV LOS formula:

$$PL_{\text{LOS,AV}} = 28 + 22\log_{10}(d_{3D}) + 20\log_{10}(f_{c,\text{GHz}}). \quad (2.23)$$

For aerial NLOS links, the implemented UMa-AV NLOS expression is defined for $22.5 < h_{\text{UE}} \leq 100$ m:

$$PL_{\text{NLOS,AV}} = -17.5 + (46 - 7\log_{10}(h_{\text{UE}}))\log_{10}(d_{3D}) + 20\log_{10}\left(\frac{40\pi f_{c,\text{GHz}}}{3}\right). \quad (2.24)$$

The path loss stored in the dataset is selected according to the cached LOS/NLOS state:

$$PL = \begin{cases} PL_{\text{LOS}}, & \text{for a low-height LOS link,} \\ PL_{\text{NLOS}}, & \text{for a low-height NLOS link,} \\ PL_{\text{LOS,AV}}, & \text{for an aerial LOS link,} \\ PL_{\text{NLOS,AV}}, & \text{for an aerial NLOS link with } 22.5 < h_{\text{UE}} \leq 100. \end{cases} \quad (2.25)$$

Thus, the logged path-loss value is not only a function of distance. It also reflects UE height, carrier frequency, the sampled LOS/NLOS state, and the UMa or UMa-AV scenario selected for the link.

2.2.4 Shadow-Fading Model

Shadow fading models slower received-power variation caused by obstacles and local environmental structure that are not captured by distance-only path loss. It is commonly represented as a log-normal random component, or equivalently as a Gaussian random variable in dB. In the implemented simulator, the shadow-fading loss for a UE–gNB link is sampled as

$$X_\sigma \sim \mathcal{N}(0, \sigma_{\text{SF}}^2), \quad (2.26)$$

where σ_{SF} is selected from the corresponding LOS/NLOS and UE-height condition. For UMa-AV links, the standard deviation follows the shadow-fading parameters in 3GPP TR 36.777 Annex B, with the low-height UMa values referenced from TR 38.901 [9, 8]. The simulator uses

$$\sigma_{\text{SF}} = \begin{cases} 4 \text{ dB}, & 1.5 \leq h_{\text{UE}} \leq 22.5 \text{ m, LOS,} \\ 6 \text{ dB}, & 1.5 \leq h_{\text{UE}} \leq 22.5 \text{ m, NLOS,} \\ 4.64e^{-0.0066h_{\text{UE}}} \text{ dB}, & 22.5 < h_{\text{UE}} \leq 300 \text{ m, LOS,} \\ 6 \text{ dB}, & 22.5 < h_{\text{UE}} \leq 100 \text{ m, NLOS.} \end{cases} \quad (2.27)$$

The implementation uses the same UMa/UMa-AV wrapper functions for all UE heights. When $h_{\text{UE}} = 1.5$ m, these functions reduce to the low-height UMa case. Therefore, the simulator does not use a separate ground-mode shadow-fading configuration; the shadow-fading standard deviation is always selected from the UE-height and LOS/NLOS conditions shown above.

The sampled value is clipped to the interval $[-3\sigma_{\text{SF}}, 3\sigma_{\text{SF}}]$. This clipping avoids unrealistically large random losses in finite simulation traces while preserving the zero-mean Gaussian behaviour for most samples. The received power including shadow fading is then

$$P_{\text{rx}} = P_{\text{tx}} + G_{\text{tx}} + G_{\text{rx}} - PL - X_\sigma, \quad (2.28)$$

where P_{tx} is the transmit power, G_{tx} and G_{rx} are antenna gains, PL is path loss, and X_σ is the shadow-fading term. A positive shadow-fading realization reduces the received power, while a negative realization improves it relative to the median path-loss

value. This explains why two UE positions with similar distance to the serving gNB may still produce different received powers.

The simulator does not resample shadow fading independently at every time step. Instead, the current shadow-fading value is cached for each UE–gNB link and reused while the accumulated UE movement for that link remains below 25 m. Once the accumulated movement reaches 25 m, a new X_σ value is sampled and the accumulated distance is reset. This 25 m update distance follows the distance-dependent shadowing treatment used in 3GPP channel modelling [9]. As a result, shadow fading changes more slowly than instantaneous RSRP or SINR calculations and avoids artificial step-to-step randomness in the generated measurement trace.

2.2.5 RSRP, Interference, Noise, and SINR

RSRP is a key measurement for assessing cell strength. In NR, 3GPP TS 38.215 defines SS-RSRP and CSI-RSRP as linear averages of the power contributions of resource elements carrying the corresponding synchronization-signal or CSI reference signals [1]. In this thesis, RSRP is used as a system-level approximation of reference-signal power derived from the received downlink power. It provides a compact indicator of serving-cell and neighboring-cell strength and is therefore useful for comparing candidate gNBs along a UE trajectory.

Interference is produced by transmissions from non-serving cells that reuse the same time-frequency resources or overlap with the measured resources. Unlike thermal noise, interference depends on the topology, the UE position, the serving gNB, and the received powers from surrounding gNBs. A UE near a cell edge, or an aerial UE with LOS visibility to several gNBs, may observe strong desired-signal power and strong interference at the same time. For this reason, RSRP alone is not sufficient to describe link quality.

Receiver noise is commonly modelled from the thermal noise density and receiver noise figure:

$$N_{\text{dBm}} = -174 + 10\log_{10}(B) + NF, \quad (2.29)$$

where B is the receiver bandwidth in Hz and NF is the receiver noise figure in dB. The value -174 dBm/Hz is the approximate thermal noise power spectral density at room temperature [7]. Noise is independent of the serving-cell choice, but it becomes important when the received signal and interference powers are low.

SINR combines the desired signal, interference, and noise into one quality metric. In linear scale, it is written as

$$\text{SINR} = \frac{P_{\text{serving}}}{\sum_{i=1}^{N_{\text{int}}} P_{\text{int},i} + P_{\text{noise}}}, \quad (2.30)$$

and in dB as

$$\text{SINR}_{\text{dB}} = 10 \log_{10}(\text{SINR}). \quad (2.31)$$

3GPP TS 38.215 also specifies NR physical-layer SINR measurements, including SS-SINR and CSI-SINR, based on reference-signal resources and corresponding interference-plus-noise contributions [1]. Conceptually, SINR is more directly related to link quality than RSRP because it penalizes both weak serving-signal power and strong interference. Thus, a UE can have acceptable RSRP but poor SINR if neighboring cells are received strongly.

The radio-measurement dataset generated in this thesis records these propagation and measurement concepts in structured form. UE position, height, velocity, and direction describe the mobility state. Serving-cell information identifies the current associated gNB. LOS probability and LOS state describe the propagation condition. Path loss, shadow fading, received power, RSRP, neighboring-cell RSRP, and SINR describe the radio environment observed by the UE at each simulation step. Together, these fields make it possible to inspect how movement changes radio conditions and how those changes may affect cell association and HO-related behavior.

2.3 Mobility Management and Handover Delay

Mobility management is the set of network functions that maintains service continuity while a UE changes its position, serving-cell condition, or radio environment. In an NG-RAN system, the UE is associated with a serving gNB, while neighboring gNBs may become suitable targets as the UE moves through the deployment area. The serving-cell choice is therefore related to radio measurements such as RSRP and SINR, but it is also constrained by network procedures, signaling, and resource preparation [2, 1].

At the radio-measurement level, mobility management starts from observing the quality of the serving cell and neighboring cells. A UE near a cell boundary may receive comparable signals from several gNBs, and an aerial UE may observe even more visible cells because of stronger LOS conditions. If a neighboring cell becomes more favorable than the current serving cell, the network may need to perform a HO to preserve link quality and avoid service degradation. This radio-level view explains why RSRP, SINR, LOS state, interference, UE position, and speed are important fields in the generated measurement trace.

HO is the mobility procedure that transfers a UE from a source cell or gNB to a

target cell or gNB while attempting to maintain service continuity. Conceptually, a handover includes target-cell selection, preparation of resources at the target side, signaling between network entities, UE reconfiguration or access to the target cell, and completion or failure handling [2]. Therefore, a change in the best radio candidate should not be interpreted as a completed handover by itself. It indicates that a mobility action may be needed, whereas the actual handover outcome depends on whether the required protocol procedure is successfully executed.

In this thesis, HO delay refers to the time interval required for the HO procedure to progress from its trigger or preparation phase to its observed completion or failure outcome in the available experiment logs. This delay is affected not only by radio conditions, but also by signaling latency, packet loss, jitter, retransmission behavior, and timer configuration. Therefore, HO-delay analysis requires protocol-level event logs in addition to radio-measurement traces. The radio simulator is used to describe serving-cell changes and measurement conditions, while the handover-emulation logs are used to analyze protocol-level completion time, timeout cases, and failure behavior.

2.4 Open-Source Platforms for 5G Core Networks

Open-source 5G core platforms provide deployable implementations of the control-plane and user-plane functions used in 5G Standalone (SA) systems. They are important for mobility research because radio measurements alone cannot show how a handover is executed after a target cell is selected. A protocol-level testbed needs core-network functions that can handle UE registration, session management, NGAP/N2 signalling, and user-plane connectivity, so that handover procedures can be observed through logs and timers instead of being treated only as abstract events [2]. In this thesis, free5GC and Open5GS are considered as representative open-source platforms.

free5GC is an open-source implementation of the 5GC architecture defined for 3GPP Release 15 and later systems [10]. It provides modular network functions such as AMF, SMF, UPF, NRF, UDM, AUSF, PCF, and NSSF, which correspond to the service-based 5GC architecture. This makes free5GC suitable for experiments that require explicit 5G SA control-plane behavior, especially AMF-side signalling over NGAP/N2 and user-plane support through UPF/GTP-U. For this reason, free5GC is the primary 5G core platform used by the handover-emulation part of this thesis.

Open5GS is another open-source mobile core platform that supports both 5GC and EPC deployments [11]. Its broader NR/LTE scope makes it useful for constructing private mobile-network testbeds and for comparing implementation choices across 5G and LTE-oriented systems. In the context of this thesis, Open5GS is treated mainly as a reference platform, while free5GC is more directly aligned with the implemented 5G

SA handover experiment.

Table 2.2 Comparison of open-source 5G core platforms

Criterion	free5GC	Open5GS
Main purpose	Open-source implementation of the 5GC based on 3GPP Release 15 and beyond [10].	Open-source implementation for both 5GC and EPC, supporting NR/LTE core-network deployment [11].
Network generation focus	Mainly focused on 5G SA core-network experiments.	Covers both 5G and LTE core-network scenarios.
Architecture	Follows the 5GC SBA with modular network functions such as AMF, SMF, UPF, NRF, UDM, AUSF, PCF, and NSSF.	Provides 5GC and EPC functions for NR/LTE networks, making it suitable for mixed-generation testbeds.
Deployment orientation	Supports cloud-native deployment with Docker and Kubernetes, which is convenient for reproducible research testbeds [10].	Provides documentation and source code for building private NR/LTE mobile-network testbeds [11].
Suitability for this thesis	Used as the main 5G core platform for protocol-level handover emulation.	Considered as a reference platform rather than the main experimental core.

This comparison clarifies why the implemented test environment uses free5GC for the 5G SA handover study, while Open5GS remains useful as a reference point for alternative open-source core-network testbeds.

2.5 Chapter Conclusion

This chapter presented the theoretical background used in the thesis, including 5G mobility, A2G communication, mobility modelling, radio propagation, RSRP, SINR, HO delay, and open-source 5G core platforms. These concepts explain the radio and protocol terms used in the rest of the report. The next chapter describes how these concepts are implemented in the system design for radio-data generation and HO-delay analysis.

CHAPTER 3. INVESTIGATION OF MOBILITY MANAGEMENT IN AERIAL-TO-GROUND COMMUNICATIONS

3.1 System Architecture

The proposed evaluation platform is organized as a measurement-driven handover analysis chain with two complementary subsystems. The radio simulation subsystem produces radio-measurement traces from a controlled mobility and propagation scenario. The handover-emulation subsystem then replays those measurements in StormSIM to execute and observe protocol-level handover behavior through a 5G Core environment.

The radio simulation subsystem is responsible for the measurement side of the study. It defines the simulation area, places gNBs according to the selected cell layout, updates UE positions according to the mobility model described later in this chapter, evaluates the propagation model, and records the radio state at each simulation step. The recorded state includes UE position and height, serving-cell information, LOS/NLOS condition, path loss, shadow fading, received power, RSRP, SINR, and neighbor-cell measurement context. This subsystem therefore answers the radio-level question: under the configured scenario, when do the measurements suggest that another gNB should become the serving cell?

The handover-emulation subsystem is responsible for the protocol side of the study. StormSIM emulates the UE and gNB behavior needed for the selected handover scenarios and interacts with the 5G Core through the corresponding control-plane and user-plane interfaces. Before handover evaluation, the UE is registered and a data session is established. After that, the measurement trace is replayed as the input that determines handover opportunities. For each handover trial, the emulator applies the configured transport conditions, executes the selected handover procedure, observes timer behavior, and records the resulting success or failure and delay.

The interface between the two subsystems is an intermediate measurement trace. It allows the radio simulator and the protocol emulator to remain independently executable while still forming one evaluation workflow. Once the trace is fixed, the same radio-generated serving-cell sequence can be replayed under different handover and transport-impairment settings. The normalization step at this boundary maps cell identifiers and measurement fields into the representation required by StormSIM; it is therefore an interface adaptation stage rather than part of the propagation model or the signaling procedure.

With this architecture, Chapter 4 can report radio-measurement behavior and protocol-level outcomes as linked but separate results: the trace provides the serving-cell se-

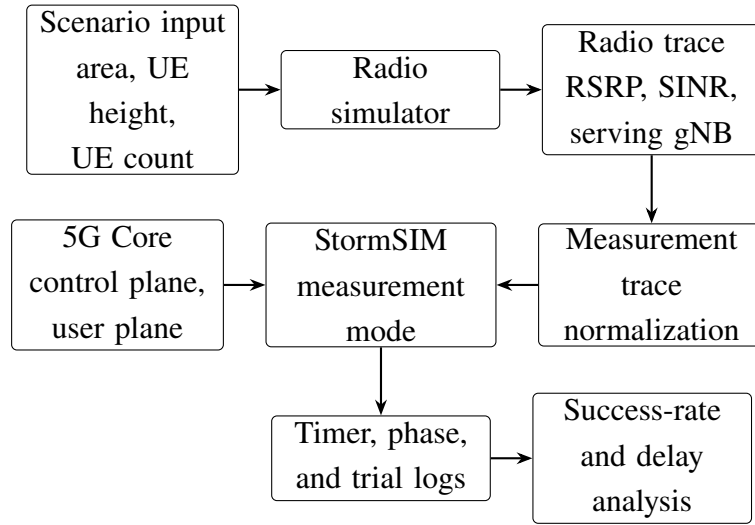


Figure 3.1 Overall architecture of the measurement-driven handover evaluation platform

Table 3.1 Functional boundary between the radio simulator and StormSIM

Component	Main responsibility	Output used in this thesis
Radio simulation subsystem	Generate mobility and radio measurements from the selected topology, UE movement, LOS/NLOS state, path loss, shadow fading, RSRP, received power, and SINR.	Measurement trace with serving-cell state, handover indication, serving-cell quality, and neighbor-cell context.
Measurement-trace interface	Adapt the radio-measurement output into the input representation required by the handover emulator.	A normalized measurement sequence used to trigger or replay handover opportunities.
StormSIM and 5G Core	Register UE, establish a data session, execute handover procedures, apply transport impairment, monitor timers and handover phases, and record trial outcomes.	Timer events, phase timestamps, trial outcomes, success rate, and delay statistics.

quence, while StormSIM provides timer events, trial outcomes, and handover-delay statistics.

3.2 Radio Simulation Design

3.2.1 Network Layout

The radio simulation subsystem is designed to generate structured radio-measurement traces from a controllable geometric scenario. The simulation area entered by the user is represented as a rectangular study region placed at the center of a 10 km by 10 km canvas. This separation between the full canvas and the active study rectangle makes it possible to draw the cellular layout around the region while keeping the UE movement bounded inside the selected area.

Ground gNBs are deployed on a flat-top hexagonal grid. The geometric hexagon radius is $R = 500$ m, so the distance between adjacent gNB centers is approximately $\sqrt{3}R$, or about 866 m. This value is not entered manually for every gNB. Instead, the simulator generates enough candidate hexagonal sites to cover the user-defined rectangle and a surrounding buffer. As a result, the number of gNBs automatically follows the size of the selected study area.

The network layout is later combined with the UE position, height, direction, and speed supplied by the mobility model in Section 3.3 to compute the UE–gNB geometry and radio measurements.

The gNB placement procedure follows the axial-coordinate representation for hexagonal grids. Each hexagon is stored as (q, r) , and the third cube coordinate is recovered as

$$s = -q - r. \quad (3.32)$$

This keeps the cube-coordinate constraint $q + r + s = 0$ while storing only two coordinates, which is a standard practical approach for hexagonal-grid algorithms [12]. The flat-top axial-to-Cartesian conversion used for the gNB layout is

$$x = R \left(\frac{3}{2}q \right), \quad (3.33)$$

$$y = R \left(\sqrt{3}r + \frac{\sqrt{3}}{2}q \right), \quad (3.34)$$

where R is the geometric hexagon radius. These coordinates are shifted by the canvas center so that the generated grid is centered on the simulation map.

To adapt the topology to an arbitrary user-defined rectangle, the algorithm first estimates the axial coordinate ranges needed to cover the requested width and height:

$$q \in \left[\text{int} \left(-\frac{W}{1.5R} \right) - 1, \text{int} \left(\frac{W}{1.5R} \right) + 1 \right], \quad (3.35)$$

$$r \in \left[\text{int} \left(-\frac{H}{\sqrt{3}R} \right) - 1, \text{int} \left(\frac{H}{\sqrt{3}R} \right) + 1 \right], \quad (3.36)$$

where W and H are the user-entered simulation width and height, and $\text{int}(\cdot)$ denotes integer conversion. The additional margin of one axial step avoids missing edge cells after the skewed axial grid is converted into Cartesian coordinates. The simulator then enumerates all (q, r) pairs in these ranges, computes $s = -q - r$, converts the hex center to Cartesian coordinates, and keeps only the centers that fall inside the simulation rectangle expanded by a boundary buffer:

$$b = \frac{\sqrt{3}}{2}R. \quad (3.37)$$

Thus a candidate gNB is retained when

$$x_{\min} - b \leq x \leq x_{\max} + b, \quad y_{\min} - b \leq y \leq y_{\max} + b. \quad (3.38)$$

This design makes the gNB count a consequence of the requested study area instead of a fixed scenario constant. A larger rectangle automatically produces more hexagonal sites, while a smaller rectangle keeps only the sites needed near that region. The buffer also ensures that UEs near the simulation boundary still observe nearby external cells, reducing artificial edge effects in RSRP, SINR, and serving-cell selection.

3.2.2 Radio Parameters

The radio parameters define the fixed link-budget and measurement settings used by the radio simulator. These values are separated from scenario variables such as UE height and mobility state. In the implemented workflow, UE height is an input selected within the research range, while the remaining radio parameters provide a common baseline for received-power, RSRP, SINR, and serving-cell selection calculations.

Table 3.2 Principal radio parameters [8, 9, 13, 14]

Parameter	Value
Carrier frequency	2 GHz
gNB transmit power	46 dBm
gNB / UE antenna gain	2 / 0 dB
System bandwidth	10 MHz
UE receiver noise figure	9 dB
gNB antenna height	25 m
UE height input range	1.5–300 m
RSRP handover margin	3 dB

The carrier frequency, gNB transmit power, and antenna gains are used in the link-budget calculation after path loss and shadow fading are evaluated. The system bandwidth and receiver noise figure determine the thermal-noise level used in the SINR calculation. The fixed gNB antenna height and the selected UE height define the three-dimensional UE–gNB geometry and the propagation height regime. The UE height is therefore not treated as a single default scenario constant; it is varied within the configured 1.5–300 m range for terrestrial, transition, and aerial cases.

The RSRP value used in this thesis is a system-level approximation derived from the received downlink power. For a 10 MHz carrier, the implementation assumes 50 resource blocks based on the E-UTRA transmission-bandwidth configuration and 12 subcarriers per resource block when mapping received power to RSRP [13, 15]. This abstraction is sufficient for generating interpretable measurement traces, but it is not intended to reproduce a full physical-layer CRS-RSRP, SS-RSRP, or CSI-RSRP measurement implementation [16, 1]. The 3 dB margin is then used by the RSRP-based serving-cell rule described later in this chapter.

3.2.3 Propagation-Model Calculation Flow

The propagation model is executed for each UE–gNB candidate link at every simulation step. Its role is to transform the current geometry and UE height into radio fields that are later used for serving-cell selection, interference calculation, SINR calculation, and dataset generation. The model follows the theoretical chain introduced in Chapter 2, but here the focus is on how the chain is organized inside the simulator workflow.

The simulator does not rely on a separate terrestrial-versus-aerial mode switch. Instead, the UE height is the controlling input for the propagation regime. A low-height UE naturally falls into the UMa height range, while elevated UEs are evaluated by the

UMa-AV expressions when their height enters the aerial range. This design keeps the radio calculation consistent across all altitude cases: the same processing chain is used, and only the height-dependent model selection changes.

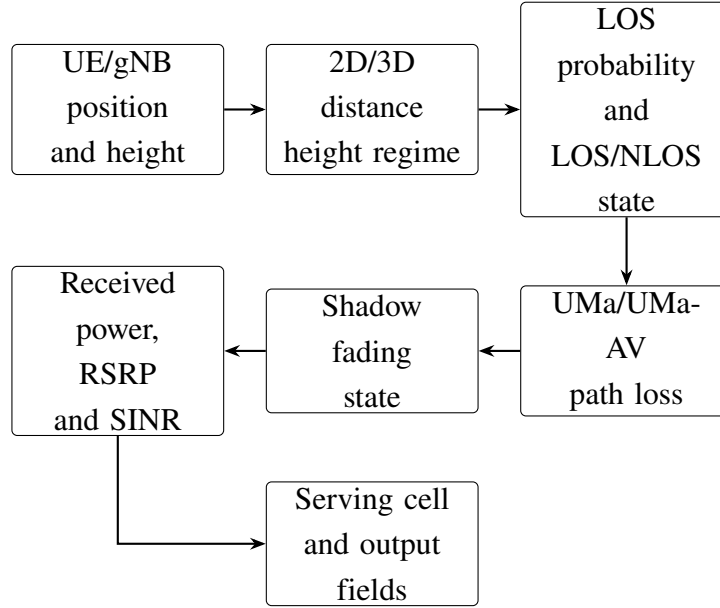


Figure 3.2 Propagation-model calculation flow in the radio simulator

This flow is applied to candidate serving cells and to the first-tier interfering cells used in the SINR calculation. For each UE–gNB link, the sampled LOS/NLOS state and the current shadow-fading value are stored in a per-link cache. The shadow-fading value is reused while the UE movement accumulated for that link remains below 25 m, and a new value is sampled after the update distance is reached. This keeps the large-scale fading variation spatially smoother than an independently sampled value at every time step.

For a selected gNB, received power and RSRP are computed as

$$P_{rx} = P_{tx} + G_{tx} + G_{rx} - PL - SF, \quad (3.39)$$

$$RSRP = P_{rx} - 10\log_{10}(50 \times 12). \quad (3.40)$$

The SINR calculation uses the serving-link received power as the desired signal. Interference is calculated from the six nearest non-serving gNBs around the serving gNB, which approximate the first tier of co-channel interferers in the hexagonal layout. Thermal noise is then added to this interference term before converting the result to dB. This design makes SINR more sensitive than RSRP to aerial scenarios, because a higher UE can observe stronger received power from both the serving gNB and neighboring gNBs. The neighbor-cell RSRP fields are therefore not only auxiliary logging fields;

they provide the radio context needed to explain why a UE may experience high serving-cell power while its SINR remains limited by interference.

3.2.4 Handover Decision and Output

In 3GPP RRC measurement reporting, Event A3 is used when a neighbouring cell becomes better than the serving cell by a configured offset, so it provides a measurement-level condition for identifying candidate serving-cell changes [17]. The radio simulator applies this idea as an RSRP-based serving-cell rule. At initialization, the strongest candidate cell is selected as the serving gNB. After that, the current serving cell is kept unless a candidate cell exceeds the current serving-cell RSRP by the configured handover margin. In the implemented radio trace generator, candidate c replaces serving cell s only when

$$\text{RSRP}_c > \text{RSRP}_s + M_{\text{HO}}, \quad M_{\text{HO}} = 3 \text{ dB}. \quad (3.41)$$

Thus, the recorded handover flag marks the simulation step where the exported serving-cell index changes according to this A3-like RSRP condition. The Python radio simulator does not apply time-to-trigger, measurement filtering, or A5 threshold checks. A5-style threshold conditions are considered in the CHO-capable StormSIM workflow when the measurement trace is replayed for protocol-level handover evaluation.

The exported dataset records UE position, height, direction, LOS probability, LOS/NLOS state, path loss, shadow fading, RSRP, received power, SINR, speed, serving gNB index, handover flag, and six neighbor-cell RSRP measurements. These fields are later used to inspect how UE movement and altitude affect the radio conditions around serving-cell changes.

3.3 UE Mobility Model

This section describes the UE mobility model used by the radio simulator. Two mobility concepts are discussed: Random Walk and Manhattan mobility. Random Walk is included as a reference model because it is a common baseline for random movement, while Manhattan mobility is the model actually used in the implemented simulation scenario. This distinction is important because the thesis studies urban-style movement where UE trajectories are constrained by road geometry rather than freely changing direction in an open plane.

The Random Walk model is an unconstrained mobility model in which the UE changes its movement direction randomly over time [18]. It is introduced only as a reference mobility concept because it allows free movement in a two-dimensional plane without road constraints. Since the implemented scenario represents urban-style move-

ment, the simulator uses the Manhattan mobility model instead.

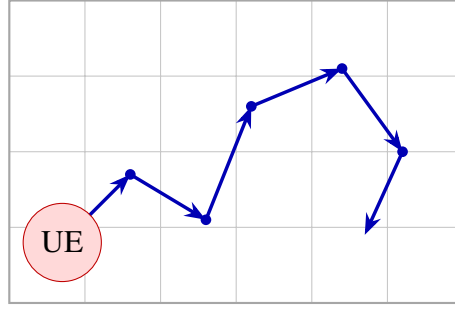


Figure 3.3 Example of UE movement in the Random Walk mobility model

The Manhattan mobility model is a geographic-restriction mobility model designed to represent movement on an urban street grid [18]. The service area is divided into horizontal and vertical road segments that form an orthogonal grid. A UE moves only along these road segments and may travel in one of four directions: East, North, West, or South. Direction changes are permitted at intersections, where the UE may continue forward or select another valid road while remaining inside the service area.

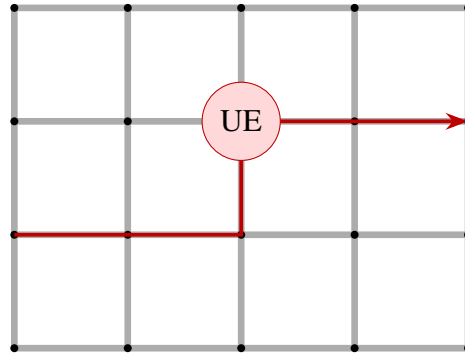


Figure 3.4 Example of UE movement in a Manhattan street grid

In the implemented simulator, the road grid is constructed from vertical and horizontal line sets over the rectangular study area. The distance between adjacent road lines is entered as a simulation parameter. Each UE begins at one of the generated road intersections, so its initial position already satisfies the Manhattan-grid constraint. At an intersection, the available movement directions are determined by the remaining road segments inside the rectangle. This prevents the UE from selecting a direction that would immediately leave the study area.

UE movement is continuous along each road segment. During one simulation interval, the nominal travel distance is determined by the base UE speed and the duration of the time step:

$$d_{\text{move}} = v\Delta t, \quad (3.42)$$

where d_{move} is the nominal distance traveled during one time step, v is the base UE speed, and Δt is the time-step duration. The base speed is sampled once for each UE from the height-dependent speed interval selected for the scenario height. Therefore, UEs in the same simulation run share the same altitude but may have different sampled speeds. This is useful for multi-UE simulations because the dataset can contain several independent trajectories under the same altitude condition.

The effective movement distance can be reduced by the stop-and-go mechanism. Let α_k denote the speed factor at time step k , where $0 \leq \alpha_k \leq 1$. The distance applied to the UE position update is

$$d_{\text{eff}}(k) = \alpha_k v \Delta t. \quad (3.43)$$

When the UE is moving normally, $\alpha_k = 1$. When the UE is stopped, $\alpha_k = 0$. During the acceleration phase after a stop, α_k increases gradually until the UE returns to its base speed. In the current simulation setting, after three direction changes the UE pauses for nine steps and then ramps up over two steps. This mechanism is not a full traffic-flow model, but it prevents the mobility trace from becoming an unrealistically constant-speed sequence and creates more varied radio-measurement behavior near intersections.

For a movement direction $u_k \in \{\text{E}, \text{N}, \text{W}, \text{S}\}$, the position update follows the road axis rather than an arbitrary angle:

$$(x_{k+1}, y_{k+1}) = \begin{cases} (x_k + d_{\text{eff}}(k), y_k), & u_k = \text{E}, \\ (x_k, y_k + d_{\text{eff}}(k)), & u_k = \text{N}, \\ (x_k - d_{\text{eff}}(k), y_k), & u_k = \text{W}, \\ (x_k, y_k - d_{\text{eff}}(k)), & u_k = \text{S}. \end{cases} \quad (3.44)$$

When the remaining distance in the current time step is larger than the distance to the next intersection, the UE first reaches that intersection, selects a valid outgoing road, and then continues moving with the remaining distance. This detail is important because it allows one simulation step to cross an intersection without losing distance, while still enforcing the Manhattan-grid constraint.

The Manhattan update procedure can be summarized as follows:

1. initialize each UE at a random road-grid intersection;
2. assign an initial valid direction from the available cardinal directions;

3. compute the effective travel distance for the current step;
4. move the UE along the current road segment until the distance is exhausted or an intersection is reached;
5. at an intersection, randomly select one valid direction and update the turn counter if the direction changes;
6. trigger a pause and ramp-up cycle after the configured number of direction changes.

This mobility model is especially relevant to the radio-measurement objective of the thesis. As the UE moves along the street grid, its distance and relative direction to surrounding gNBs change over time. These changes affect LOS probability, path loss, shadow fading, RSRP, interference, and SINR. When a neighboring gNB becomes stronger than the current serving gNB by more than the handover margin, the serving-cell index changes and a handover flag is recorded in the dataset. Thus, the combination of the hexagonal gNB layout and Manhattan mobility provides a consistent geometric foundation for studying radio-measurement variation and HO-related serving-cell behavior.

3.4 Implementation of Handover Procedure in StormSIM

StormSIM is used to execute the protocol-level part of the handover study. In the preceding radio simulator, a serving-cell change only indicates that the radio measurements prefer another gNB. In StormSIM, the same change is replayed as a handover opportunity so that signaling, timers, transport impairment, and completion delay can be observed. This separation follows the NG-RAN mobility view in which handover includes target preparation, UE reconfiguration or execution, path switching, and completion or failure handling, not only radio target selection [2, 17, 19].

Figure 3.5 summarizes the implemented measurement-driven handover flow. Before the handover experiment starts, StormSIM connects to the 5G Core, registers the UE, and establishes a PDU session. Packet loss is temporarily disabled during this preparation phase so that registration and session establishment do not dominate the handover results. After the UE is ready, the configured packet loss, latency, and jitter values are restored and applied to the virtual radio link and Xn path.

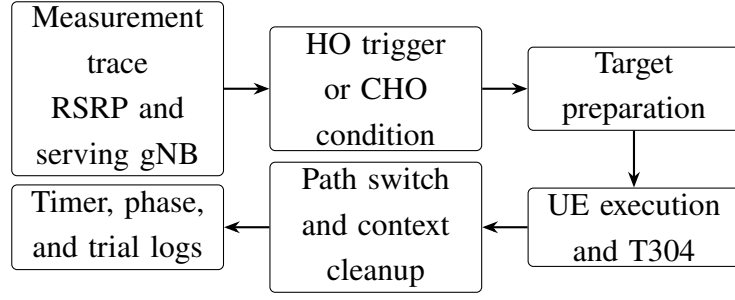


Figure 3.5 Measurement-driven handover procedure implemented in StormSIM

3.4.1 *Measurement-Driven Trigger*

The measurement input used by StormSIM contains Step, connected_gnb, neighbor-cell RSRP columns named gnbX_rsrp, and an optional Type field. The initial serving gNB is selected from the strongest RSRP value in the first measurement row. In normal measurement mode, StormSIM does not recompute the target cell from RSRP. Instead, it treats a change in connected_gnb between two consecutive rows as the handover trigger. The previous serving gNB becomes the source, the new connected_gnb becomes the target, and the Type field selects Xn or N2 handover. If the field is missing, Xn handover is used.

This design makes StormSIM a protocol emulator for an already generated measurement trace. The RSRP values remain useful as radio context and, in CHO mode, as the input used to build RRC-style measurement reports. In this mode, StormSIM smooths RSRP using a nearest-three-step window, updates the UE measurement store, and sends a MeasurementReport from the UE to the source gNB. The CHO policy then evaluates A3 and A5 conditions. A3 is satisfied when a neighbor is better than the serving cell by the configured offset and hysteresis, while A5 is satisfied when the serving cell is below a threshold and the neighbor is above a threshold [17]. The implemented default CHO parameters are a 5 dB A3 offset, 0 dB hysteresis, and -100 dBm serving and neighbor thresholds for A5.

3.4.2 *Xn, N2, and CHO Execution*

For Xn handover, the source gNB creates a target-side virtual radio connection, starts TXnRELOCprep and TXnRELOCoverall, and forwards the UE context to the target gNB. The context-forwarding message is retried up to three times with a short delay between attempts. If the forwarding step fails, StormSIM records a transport-failure event and the handover attempt is later reported as failed. If the target preparation succeeds, the source sends a handover command to the UE. The UE starts T304, executes the target connection, switches its serving gNB when the execution succeeds, and stops T304. After the target gNB obtains the UE context, it performs path switching toward the core

network; the source-side Xn relocation timers are stopped when the successful execution and path-switch stages are observed.

For N2 handover, the source gNB sends an NGAP HandoverRequired message toward the AMF and starts TNGRELOCprep and TNGRELOCoverall. The target preparation, handover command, and path switch are then handled through the NGAP procedure between the gNBs and the 5G Core [19]. In the thesis workflow, the Type field in the measurement trace selects whether the triggered attempt follows the Xn or N2 branch.

StormSIM also includes a CHO-capable execution path. In this path, the source gNB prepares candidate target gNBs and sends a CHOConfig to the UE. The UE evaluates the stored candidate conditions every 500 ms using the latest RSRP values. When the condition is satisfied, the UE triggers execution toward the selected candidate. This is different from normal measurement mode: normal mode replays serving-cell changes that already exist in the trace, while CHO mode uses measurement reports and candidate conditions before executing the handover.

3.4.3 Timer and Monitor Output

The handover monitor records the high-level procedure state as

$$\text{NULL} \rightarrow \text{PREPARE} \rightarrow \text{EXECUTE} \rightarrow \text{COMPLETE} \rightarrow \{\text{SUCCESS}, \text{FAIL}\}. \quad (3.45)$$

The implemented timer configuration used by the current StormSIM source is listed in Table 3.3. These timers are used to separate preparation failure, execution failure, recovery behavior, and overall procedure completion.

Table 3.3 Implemented handover timers in StormSIM

Timer	Value	Timer	Value
TXnRELOCprep	3 s	TXnRELOCoverall	6 s
TNGRELOCprep	5 s	TNGRELOCoverall	10 s
T304	1000 ms	T310	1 s
T311	3 s	T301	2 s
T312	500 ms	T430	30 s

The timer values in Table 3.3 are the default values used by the current StormSIM source, not parameters read from the YAML scenario file. Most of them are hard-coded as timer constants in the timer package. T304 is defined as a 1000 ms default variable and is changed only if the measurement scenario explicitly overrides the

T304DurationMs field at startup.

StormSIM exports three main handover-related outputs. The `timer-events-*.csv` file stores timer start, stop, timeout, cancel, restart, and transport-failure events. The `handover-trial-summary-*.csv` file stores one row per handover attempt, including trial identifier, source gNB, target gNB, handover type, final result, failure reason, and total handover time. The `ho-time-log.csv` file stores timestamps for the monitor phases and is used to inspect the order of preparation, execution, and completion events.

3.4.4 Evaluation Metrics

The primary metrics are handover success rate, failure rate, and handover completion time. For N_{HO} attempted handovers and N_{succ} successful handovers, the success rate is

$$R_{\text{succ}} = \frac{N_{\text{succ}}}{N_{\text{HO}}} \times 100\%. \quad (3.46)$$

For successful attempts, the delay is measured from the moment StormSIM starts the handover procedure to the moment the target gNB reports successful completion. The reported delay statistics include mean, median, and 95th-percentile delay. Failed attempts are analyzed using the failure reason and timer events, such as Xn transport failure, preparation timeout, T304 timeout, or recovery-timer behavior.

Two additional mobility-quality indicators can be derived from the same logs. A ping-pong handover is counted when the UE moves from gNB *A* to gNB *B* and then returns to *A* within a short observation window. This metric is important because frequent back-and-forth handovers indicate unstable target selection near cell boundaries. Radio-link-failure-related behavior is identified from T310, T311, T301, and T312 timer events and from failed trials that do not complete before the configured timeout. In the exported logs, the column named `pdr` is populated from the configured packet-loss value; therefore, it is interpreted in this thesis as loss probability rather than as packet delivery ratio.

3.5 Chapter Conclusion

This chapter described the system design used in the thesis. It presented the radio simulation workflow, the handover-emulation workflow, the measurement-driven trigger mechanism, the recorded output fields, and the metrics used for delay and success analysis. The design also separates radio-level serving-cell behavior from protocol-level handover execution, which is necessary for interpreting the results correctly. The next chapter uses the available generated data and experiment logs to evaluate radio behavior, handover success, delay, timer behavior, and current experimental limitations.

CHAPTER 4. EXPERIMENTAL RESULTS

4.1 Experimental Scenarios

4.1.1 *Experiment Objectives*

The first objective is to evaluate the effect of UE/UAV altitude on the generated radio measurements. A higher aerial UE may have a higher probability of LOS connection to several gNBs, which can improve the serving-link received power but can also increase interference from neighboring cells. Therefore, altitude is expected to affect RSRP and SINR in different ways.

The second objective is to evaluate mobility and handover behavior. The simulator records the serving BS, neighboring-cell RSRP values, UE speed setting, and handover indicators. These outputs make it possible to identify when the serving BS changes, where handover events occur along the trajectory, and whether the number of handovers changes across altitude ranges.

The third objective is to evaluate protocol-level handover delay under selected delay and loss settings. This part uses handover-emulation logs to measure completion time, timeout behavior, and failure cases. The purpose is not to merge radio measurement and protocol delay into one metric, but to compare the radio-level trigger behavior with the protocol-level execution behavior.

4.1.2 *Altitude-Range Design*

The altitude cases are grouped by height intervals rather than by a single fixed UAV height. This design follows the height regimes used by the propagation models in Chapter 2 and Chapter 3. In the 3GPP UMa model, low-height terrestrial UE modelling is used for $1.5 \leq h_{\text{UE}} \leq 22.5$ m. The height-correction term in the LOS-probability expression becomes active above 13 m. For aerial vehicles, 3GPP TR 36.777 extends the UMa model to UMa-AV conditions for $22.5 < h_{\text{UE}} \leq 300$ m, with the implemented aerial NLOS expression used up to 100 m and a LOS-dominant high-altitude branch above 100 m [8, 9].

Table 4.1 Altitude ranges used for A2G radio-measurement experiments

Range	Height interval	Model interpretation
H0	$1.5 \leq h_{\text{UE}} \leq 13 \text{ m}$	Ground or low terrestrial UE range under the UMa model.
H1	$13 < h_{\text{UE}} \leq 22.5 \text{ m}$	Elevated terrestrial or low-height UE range under the UMa model.
H2	$22.5 < h_{\text{UE}} \leq 100 \text{ m}$	Low-to-medium aerial UE range under the UMa-AV model.
H3	$100 < h_{\text{UE}} \leq 300 \text{ m}$	High aerial UE range under the UMa-AV model.

4.1.3 Scenario Description

Table 4.2 Experimental scenario groups

Scenario	Description	Purpose
S1	Baseline terrestrial and low-height UE scenario using H0 and H1.	Provide a reference for comparing aerial UE radio behavior.
S2	A2G scenario with different altitude ranges using H2 and H3.	Analyze the impact of UAV altitude on RSRP, SINR, shadow fading, serving-cell changes, and handover indicators.
S3	A2G scenario with different handover-delay or transport-delay settings.	Analyze the impact of delay on handover completion time, timeout cases, and failure behavior.

The radio simulation settings are kept fixed across altitude scenarios except for UE height. The fixed radio parameters follow the implementation settings summarized in Table 3.2. This makes the altitude effect easier to interpret because the carrier frequency, gNB transmit power, antenna gains, bandwidth, noise figure, handover margin, and mobility model remain unchanged across the compared altitude groups.

Table 4.3 Experiment controls for A2G simulation scenarios

Control item	Value	Role in the experiment
UE height	Ranges H0–H3	Main variable in altitude scenarios
Mobility model	Manhattan mobility	UE moves along an urban road grid
UE speed	Fixed within each trace	Selected from the configured height-dependent range before each run
Simulation duration	301 samples per trace	Output steps indexed from 0 to 300 in the generated CSV files
Number of UEs	1 in the radio-measurement traces	Provides one trajectory per generated CSV file
Repetitions	51–52 traces per altitude range	Reduces randomness from mobility, LOS/NLOS sampling, and shadow fading

For statistically meaningful comparison, each altitude range is evaluated using independently generated UE trajectories. The available radio-measurement dataset contains 52 traces for H0 and 51 traces for each of H1, H2, and H3. These repetitions reduce the effect of random mobility, LOS/NLOS sampling, and shadow fading on the reported altitude comparison.

4.1.4 Evaluation Metrics

The radio-level metrics are extracted from the simulator CSV output. RSRP is used to evaluate serving-cell and neighboring-cell signal strength. SINR is used to evaluate link quality under interference and thermal noise. Shadow fading is used to observe slow large-scale signal variation around the median path loss. UE speed is treated as a fixed input within each generated trace rather than as a plotted result metric. The serving BS index and handover flag are used to identify cell changes and handover events.

For each scenario, the evaluation includes both time-series and scenario-level statistics. Time-series plots show how RSRP, SINR, and shadow fading evolve along representative UE trajectories. Scenario-level plots compare the distribution and average value of RSRP and SINR across altitude ranges. Handover-related metrics are analyzed separately using serving-cell changes and protocol-level logs when available.

4.2 Radio Measurement Results

This subsection analyzes the radio-measurement traces generated by the simulator for the four altitude ranges defined in Table 4.1. The analysis uses the CSV files stored

in the four height-range folders. The available dataset contains 52 traces for H0 and 51 traces for each of H1, H2, and H3. Each trace contains 301 time samples for one UE trajectory. The metrics used in this subsection are RSRP, SINR, and shadow fading. UE velocity is not analyzed here because each generated trace uses a fixed speed setting, so velocity mainly acts as an input condition rather than a result metric.

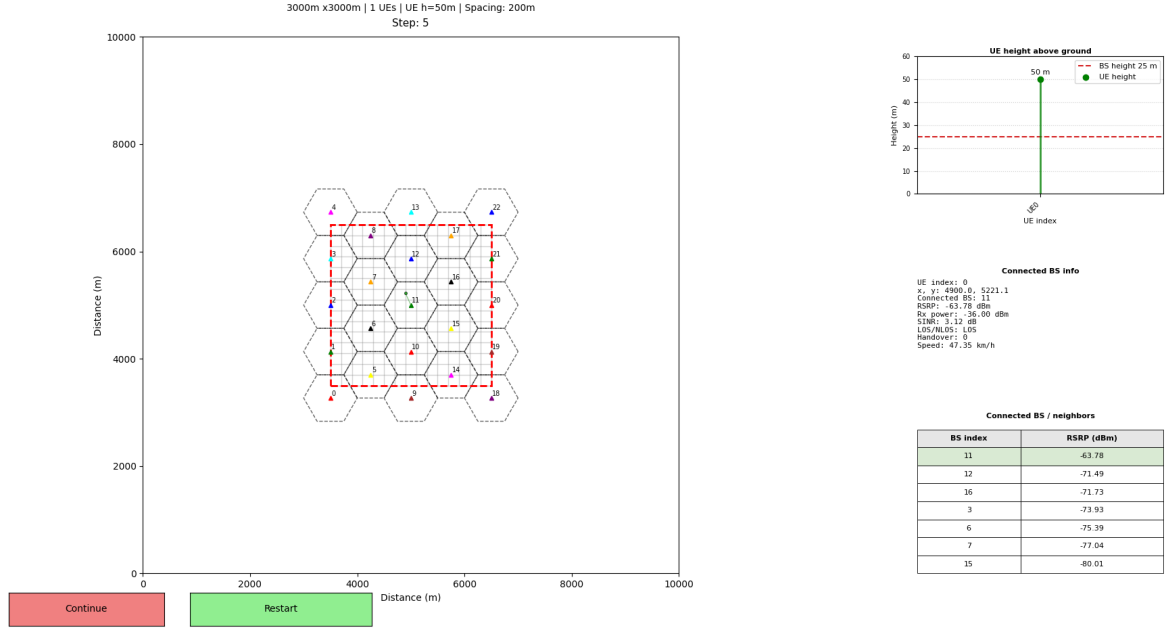


Figure 4.1 Radio-data simulation interface for a single-UE scenario

Figures 4.1 and 4.2 show the simulator interface used to generate radio-measurement data. These screenshots use an active simulation region of $3000\text{m} \times 3000\text{m}$, a UE height of 50 m, and a Manhattan road-grid spacing of 200 m. The red dashed rectangle marks the active movement region inside the larger drawing canvas, while the hexagonal layout represents the deployed gNB sites around the region. The first screenshot shows the one-UE case used for trace-level inspection, and the second screenshot shows the same scenario configuration with five UEs to verify that the simulator can generate multiple simultaneous mobility and radio-measurement records.

Table 4.4 summarizes the measured values across all CSV samples in each altitude range. The terrestrial ranges H0 and H1 have weaker average serving-cell RSRP than the aerial ranges. The mean RSRP improves from -85.46 dBm in H0 and -78.95 dBm in H1 to -68.34 dBm in H2. This follows the expected A2G trend: when the UE height increases, the link has a higher chance of favorable propagation and the serving gNB can often be received with higher power.

Figure 4.3 shows representative RSRP and SINR time series. The representative trace for each altitude range is selected as the trace whose mean RSRP is closest to the median trace-level RSRP of that range. In the RSRP curves, local decreases appear when

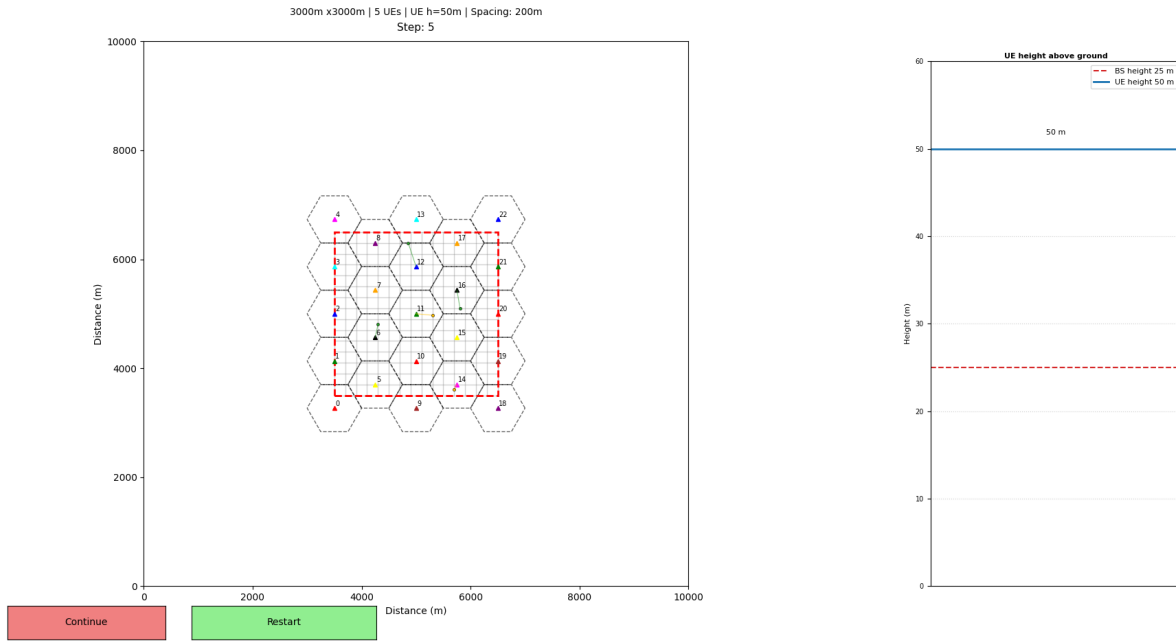


Figure 4.2 Radio-data simulation interface for a five-UE scenario

Table 4.4 Summary of radio-measurement results across altitude ranges

Range	Height inter- val	Traces	Samples	RSRP (dBm)	SINR (dB)	Shadow fading (dB)
H0	1.5–13 m	52	15652	-85.46 ± 10.92	14.59 ± 11.76	-1.83 ± 5.26
H1	13–22.5 m	51	15351	-78.95 ± 9.71	11.66 ± 9.78	-1.78 ± 5.21
H2	22.5–100 m	51	15351	-68.34 ± 5.03	2.64 ± 5.47	-0.92 ± 3.28
H3	100–300 m	51	15351	-69.92 ± 3.32	0.31 ± 3.56	-0.12 ± 1.45

the UE moves farther from the currently serving gNB or enters a less favorable propagation condition. RSRP may recover after the serving-cell selection changes to a stronger gNB. Therefore, the RSRP curve reflects both geometric movement and serving-cell changes.

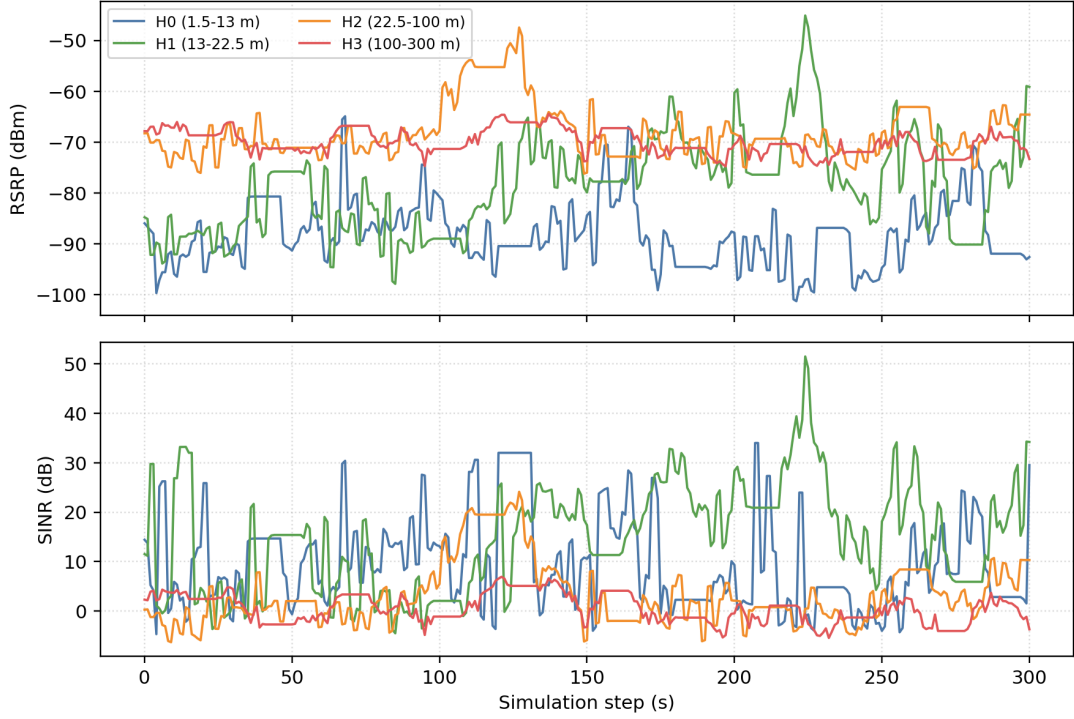


Figure 4.3 Representative RSRP and SINR time series across altitude ranges

The SINR curves show a different trend from RSRP. Although H2 and H3 have stronger average RSRP than H0 and H1, their average SINR is much lower. In Table 4.4, the mean SINR decreases from 14.59 dB in H0 and 11.66 dB in H1 to 2.64 dB in H2 and 0.31 dB in H3. This means that stronger received power at higher altitude does not automatically imply better link quality. At aerial heights, the UE can observe stronger signals not only from the serving gNB but also from neighboring gNBs. The increased neighboring-cell visibility raises the interference level, so SINR can decrease even when serving-cell RSRP remains high. This behavior is especially visible near cell-edge regions, where the serving and neighboring gNB powers become comparable.

Figure 4.4 shows the shadow-fading value of the serving link for representative traces. Shadow fading creates slow fluctuations around the median path loss and can change the received power even when the UE movement is smooth. The fluctuation range is larger in H0 and H1 than in H2 and H3. This is also reflected in Table 4.4, where the shadow-fading standard deviation is about 5.26 dB in H0 and 5.21 dB in H1, but decreases to 3.28 dB in H2 and 1.45 dB in H3. The high-altitude range therefore produces a more stable shadow-fading component in the generated traces.

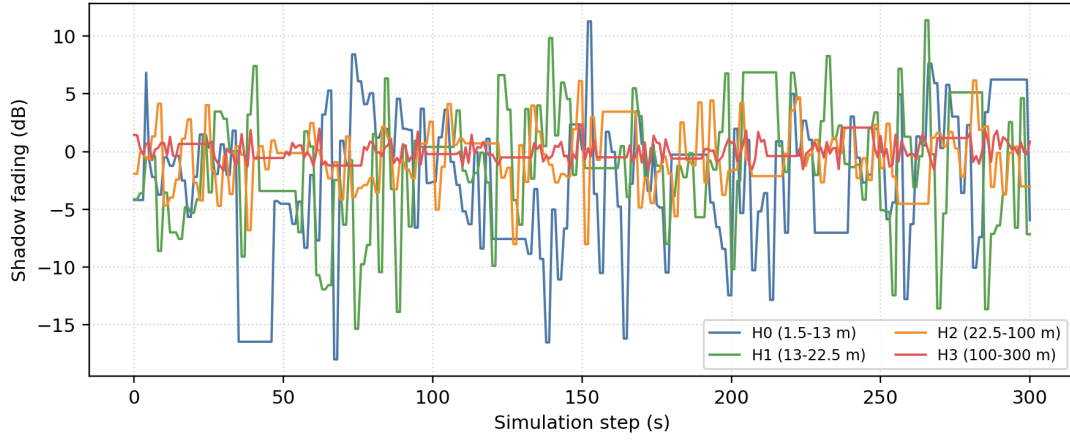


Figure 4.4 Representative shadow-fading time series across altitude ranges

The distribution-level comparison in Figure 4.5 confirms the same tendency across all traces instead of only one representative trajectory. The RSRP distribution shifts upward from the terrestrial ranges to the aerial ranges, while the SINR distribution shifts downward. The shadow-fading distribution also becomes narrower as the UE height increases, especially in H3. This confirms that altitude changes the balance between serving-link strength and interference: aerial UEs can obtain stronger received power, but they also experience stronger co-channel interference from neighboring gNBs.

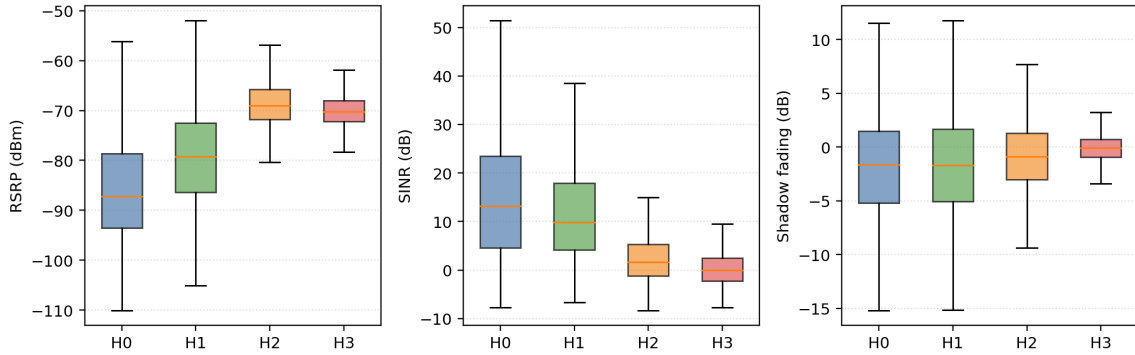


Figure 4.5 Distribution of RSRP, SINR, and shadow fading across altitude ranges

Figure 4.6 summarizes the mean RSRP and SINR values with standard-deviation error bars. The comparison is plotted with point markers instead of zero-baseline bars because RSRP is measured in negative dBm values; drawing RSRP as bars from 0 dBm would visually exaggerate the column length without adding useful information. The result highlights the main radio-measurement conclusion of this subsection: increasing altitude improves the average serving-cell RSRP up to the low-to-medium aerial range, but it does not improve SINR. The SINR reduction in H2 and H3 indicates that interference becomes the dominant limiting factor for aerial UE links in the simulated hexagonal deployment. Consequently, RSRP and SINR must be interpreted together when evalu-

ating A2G mobility. A handover decision based only on received power may identify a strong serving cell, but the resulting link quality can still be limited if several neighboring gNBs are also received strongly.

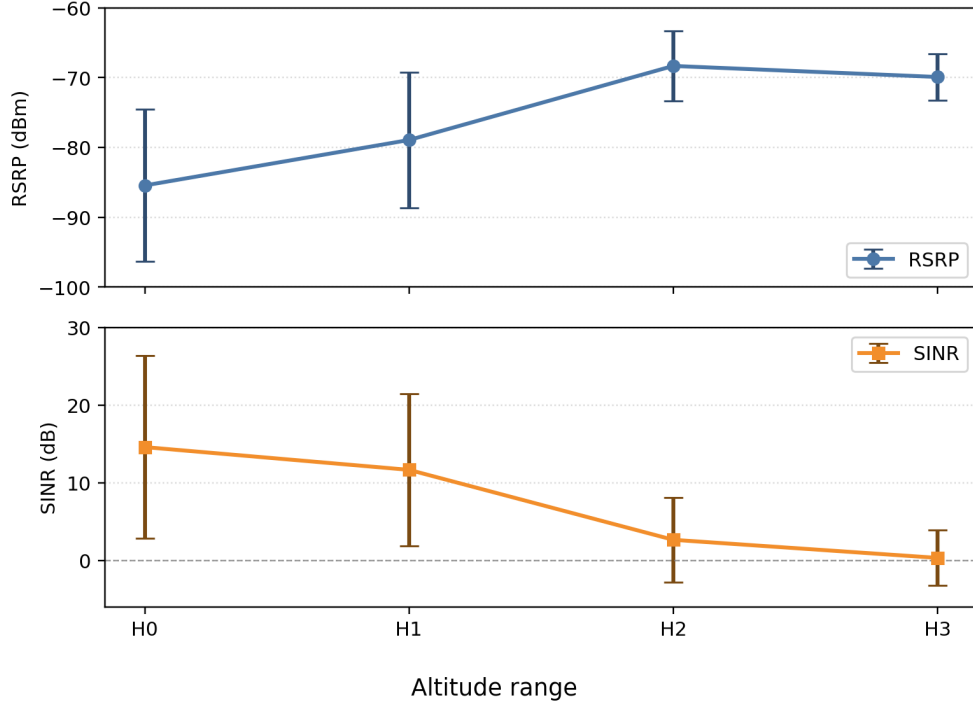


Figure 4.6 Mean RSRP and SINR comparison across altitude ranges with standard-deviation error bars

Overall, the radio-measurement results are consistent with the modelling objective of the simulator. RSRP mainly follows serving-link received power and improves when the UE has more favorable A2G propagation. SINR depends on both serving power and neighboring-cell interference, so it can degrade at higher altitude. Shadow fading introduces slow signal variation and explains why the measured radio curves are not smooth functions of distance alone.

4.3 Handover Behavior Analysis

The radio simulator identifies handover events from serving-BS changes. A handover indicator is recorded when the selected serving BS differs from the previous serving BS. This radio-level handover event should be interpreted as a serving-cell transition suggested by the measurement model, not as a completed protocol-level handover procedure.

The simulator output records the serving BS over time, the handover event timeline, the source and target BS of each serving-cell transition, and the handover count for each altitude range. These fields provide the radio-level context needed to relate serving-cell changes to the RSRP and SINR behavior reported above.

Handover events are expected to occur mainly when the UE moves toward the boundary between two cells. In this region, the RSRP of a neighboring gNB may become higher than the RSRP of the current serving gNB by more than the configured handover margin. A higher speed can make these transitions occur over fewer simulation steps, while a higher altitude can change the number of visible candidate gNBs and the interference condition. Therefore, the handover analysis should be interpreted together with the RSRP and SINR plots rather than from the handover count alone.

For protocol-level delay analysis, the handover-emulation results are evaluated separately using trial-summary and timer-event logs. The main metrics are success rate, failure rate, timeout cases, mean successful handover delay, median successful handover delay, and 95th-percentile successful handover delay. This separation preserves the distinction between a radio-level serving-cell change and the completion of the actual handover signaling procedure.

4.4 Measurement-Driven Handover-Delay Experiment

The protocol-level handover experiment uses a dedicated seven-gNB measurement trace generated from the radio simulator and replayed by StormSim. The trace topology contains one center gNB and the six first-tier neighbouring gNBs of the hexagonal layout. With the simulator hexagon radius $R = 500$ m, the neighbour spacing is $1.5R = 750$ m in the horizontal direction and $\sqrt{3}R \approx 866$ m in the vertical hexagonal direction. The active rectangular region is set to 1500 m by 1500 m. Under the implemented topology generator, this region and its boundary buffer keep exactly the center cell and the first-tier neighbours, while excluding second-tier gNBs.

The generated CSV file keeps fixed RSRP columns gnb0_rsrp to gnb6_rsrp. The index gnb0 represents the center gNB, and gnb1 to gnb6 represent the surrounding cells in angular order. The connected_gnb field uses the StormSim configuration identifiers 000000 to 000006, so that a serving-cell change in the radio trace can be replayed directly as a measurement-driven Xn handover opportunity. The StormSim configuration still contains 000007, but this gNB is not present in the measurement trace and is not selected as a handover target in this experiment.

Table 4.5 Seven-gNB StormSim measurement-trace settings

Item	Value
Measurement file	data/stormsim_7gnb_measurement.csv
Generator script	generate_stormsim_7gnb_data.py
Topology	1 center gNB and 6 first-tier neighbouring gNBs
Simulation rectangle	1500 m $\times$ 1500 m
Hexagon radius	500 m
UE height	50 m, corresponding to the H2 aerial range
Samples	301 steps, indexed from 0 to 300
Serving-cell rule	RSRP-based selection with 3 dB handover margin
Detected serving-cell changes	18 handover events
Handover type	Xn handover
Transport impairment	0 packet loss, 0 ms latency, 0 ms jitter

The experiment was executed with the following StormSim command:

```
./bin/emulator -c config/config.yml \
--csv-measurement data/stormsim_7gnb_measurement.csv \
--step-delay 10
```

Listing 1: StormSim command for the seven-gNB measurement trace

The reduced step delay is used only to replay the 301 measurement samples faster. The handover completion time is taken from StormSim trial summaries and timer logs, not from the CSV sample interval.

4.5 Handover Success and Delay

Table 4.6 summarizes the handover outcomes from the generated trial-summary log. All 18 detected serving-cell changes were executed as Xn handovers and completed successfully. No timeout or failure was recorded in this baseline run.

Table 4.6 Seven-gNB StormSim handover outcome summary

Run	Type	Trials	Success	Failure	Success rate
1782627345	Xn	18	18	0	100.0%

The successful handover delay statistics are shown in Table 4.7. The mean delay is 200.530 ms and the median delay is 200.550 ms. The minimum and maximum successful completion times are 200.122 ms and 201.013 ms, respectively.

Table 4.7 Successful Xn handover delay in the seven-gNB StormSim run

Type	Mean	Median	Minimum	Maximum	95th percentile
Xn	200.530 ms	200.550 ms	200.122 ms	201.013 ms	201.013 ms

The successful handover delays are tightly clustered around 200 ms. This does not mean that every internal handover phase takes 200 ms. Instead, the scenario waits for handover completion using a 200 ms polling ticker, so the exported trial-summary delay is quantized by the observation loop. This value is therefore suitable for comparing trial-level completion behavior in the implemented emulator, but it should not be interpreted as a precise lower-layer radio execution latency.

4.6 Timer-Level Observation

The timer-event log provides a finer view of the internal handover procedure. Table 4.8 reports successful terminal timer events from the same run. Each of the 18 Xn handover attempts produced matching start and stop records for TXnRELOCprep, TXnRELOCoverall, and T304. No timeout event was recorded.

Table 4.8 Successful timer events in the seven-gNB StormSim run

Timer	Count	Mean	Median	Range
TXnRELOCprep	18	0.389 ms	0.370 ms	0.220–0.600 ms
TXnRELOCoverall	18	0.376 ms	0.345 ms	0.200–0.620 ms
T304	18	10.621 ms	10.765 ms	10.070–10.960 ms

The timer-level data shows that internal successful timer stops occur much earlier than the 200 ms trial-summary delay. In particular, T304 stops after about 10.6 ms on average, while the Xn relocation preparation and overall timers stop within about 1 ms. The difference between these timer values and the trial-summary delay is caused by the scenario-level success polling interval. This confirms that the timer-event log and the trial-summary log should be interpreted at different granularities: timer events describe internal procedure phases, while trial summaries describe when the scenario observes the completed handover.

4.7 Chapter Conclusion

This chapter defined the experimental scenario groups for radio-measurement and handover-delay analysis. The altitude design follows the height regimes used by the UMa and UMa-AV propagation models, so the A2G experiments can compare terrestrial, tran-

sition, low-to-medium aerial, and high-aerial UE behavior. The radio-measurement results show that aerial altitude improves serving-cell RSRP but can reduce SINR because neighbouring-cell interference becomes stronger. The seven-gNB StormSim run then demonstrates the measurement-driven handover workflow: 18 Xn handovers were triggered from the generated trace and all completed successfully in the baseline no-loss condition. The trial-summary delay is clustered around 200 ms because completion is observed by a 200 ms polling loop, while the timer log shows shorter internal timer durations. The next chapter summarizes the thesis, states the main limitations, and outlines future work.

CHAPTER 5. CONCLUSION

5.1 Summary and Contributions

This thesis developed a workflow for studying 5G mobility from two complementary viewpoints: radio-level measurement behavior and protocol-level handover execution. The Python radio simulator generates structured mobility and radio-measurement traces, including UE position, UE height, serving-cell information, RSRP, SINR, shadow fading, and handover indicators. The StormSim-based emulation workflow then replays measurement-driven serving-cell changes in a free5GC environment and records handover trial summaries and timer events.

The main contribution of the thesis is the separation between a radio-level serving-cell change and a completed protocol-level handover. In the radio simulator, a handover indicator means that the selected serving gNB changes according to an RSRP-based rule. In StormSim, the handover is treated as a protocol procedure involving source gNB, target gNB, UE context transfer, path switch, T304, and Xn relocation timers. This separation avoids treating a measurement decision as if it were already a completed handover.

A second contribution is the construction of a structured radio-data generation pipeline for terrestrial and aerial UE-height ranges. The simulator supports UMa and UMa-AV propagation behavior, height-dependent mobility settings, hexagonal gNB placement, and CSV output suitable for post-processing. A third contribution is the creation of a StormSim-ready seven-gNB measurement trace with fixed gNB-indexed RSRP columns, allowing the generated radio data to be replayed by the handover emulator without losing the identity of each gNB.

5.2 Main Findings

The radio-measurement results show that UE height has different effects on RSRP and SINR. Compared with the terrestrial height ranges, the aerial height ranges produce stronger average serving-cell RSRP because the UE has more favorable propagation and a higher probability of LOS connection. However, the SINR does not improve in the same way. At aerial heights, the UE can also receive stronger signals from neighbouring gNBs, so co-channel interference increases. As a result, the average SINR decreases in the H2 and H3 altitude ranges even though the serving-cell RSRP is higher. This confirms that RSRP alone is not sufficient for evaluating A2G mobility quality.

The generated radio traces also show that shadow fading introduces slow signal variation around the path-loss trend. The shadow-fading distribution becomes narrower at higher altitude, especially in the H3 range. This behavior is consistent with the im-

plemented height-dependent propagation model and helps explain why aerial traces can have stronger but more interference-limited received signals.

For protocol-level evaluation, a seven-gNB measurement trace was generated with one center gNB and six surrounding first-tier gNBs. The trace contains 301 measurement steps and 18 serving-cell changes. When replayed in StormSim as Xn handover opportunities under the baseline no-loss condition, all 18 handover trials completed successfully. The successful trial-summary delay has a mean of 200.530 ms, a median of 200.550 ms, and a 95th percentile of 201.013 ms.

The timer-event log provides a finer interpretation of this delay. In the same run, T304 stopped successfully for all 18 handovers with an average elapsed time of 10.621 ms, while TXnRELOCprep and TXnRELOCoverall stopped within about 1 ms on average. Therefore, the approximately 200 ms trial-summary delay mainly reflects the scenario-level polling interval used to observe handover completion, not the duration of every internal protocol phase.

5.3 Limitations

The radio simulator is a controlled system-level model rather than a full physical-layer implementation. RSRP is derived from received power using a resource-grid approximation, and the serving-cell decision uses an RSRP margin without time-to-trigger, measurement filtering, or full RRC measurement-event processing. The LOS/NLOS state and shadow fading are cached per UE–gNB link to keep the trace spatially stable, but this remains an abstraction of real propagation dynamics.

The current StormSim result is a baseline validation of the measurement-driven workflow, not a complete performance study under all network conditions. The verified seven-gNB run uses Xn handover with zero packet loss, zero added latency, and zero jitter. It demonstrates that the generated measurement trace can drive successful protocol-level handovers, but it does not yet quantify handover robustness under degraded transport conditions.

The trial-summary delay is also affected by the implementation of the scenario runner. Completion is checked using a 200 ms polling ticker, so the exported successful handover duration is quantized around 200 ms. Timer-event logs are therefore necessary when interpreting lower-level handover phases. In addition, the CSV interface between the Python simulator and StormSim still requires a dedicated adapter script for fixed gNB-indexed RSRP columns.

Finally, the present evaluation focuses on Xn handover. N2 handover, full CHO mode evaluation, RLF behavior, beam-failure behavior, and systematic loss–delay experiments remain outside the verified result set reported in this thesis.

5.4 Future Work

Future work should first unify the radio-simulator and StormSim CSV formats so that fixed gNB-indexed RSRP fields are produced directly by the main simulator logger. This would remove the need for a separate adapter script and make measurement-driven handover experiments easier to reproduce.

The handover-delay evaluation should then be extended from the baseline no-loss case to a controlled matrix of packet loss, latency, and jitter settings. This matrix should report success rate, failure reason, timeout behavior, mean delay, median delay, and percentile delay for each setting. Xn and N2 handover should also be compared under the same measurement trace so that the effect of direct gNB-to-gNB handover and core-network-assisted handover can be separated.

Another important direction is to reduce or replace the 200 ms polling interval in the scenario runner. Event-driven completion logging would allow the trial-summary delay to better match the internal handover completion time. The timer lifecycle should also be checked so that each handover trial has complete and consistent start and terminal timer records.

On the radio side, future work can add time-to-trigger, measurement filtering, and additional RRC event conditions such as A5 to make the serving-cell and handover-trigger logic closer to real mobility procedures. The generated dataset can also be expanded to multiple UE trajectories and repeated fixed-seed scenarios to support more systematic statistical comparison.

5.5 Chapter Conclusion

This chapter summarized the thesis contributions, main findings, limitations, and future research directions. Overall, the thesis provides a practical platform for generating 5G mobility traces and analyzing handover behavior beyond simple serving-cell changes. The current results show that aerial UE height can improve RSRP while reducing SINR due to interference, and that a generated seven-gNB trace can drive successful Xn handovers in StormSim under baseline no-loss conditions. Broader claims about handover robustness require the controlled network-impairment and handover-type experiments described as future work.

REFERENCES

- [1] 3GPP, “NR; Physical layer measurements,” 3rd Generation Partnership Project, Tech. Rep. TS 38.215 V19.2.0, 2025. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.215/
- [2] —, “NR; NR and NG-RAN Overall Description; Stage-2,” 3rd Generation Partnership Project, Tech. Rep. TS 38.300 V19.1.0, 2026. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.300/
- [3] B. Meixner, J. W. Kleinrouweler, and P. Cesar, “4G/LTE Channel Quality Reference Signal Trace Data Set,” in *Proceedings of the 9th ACM Multimedia Systems Conference*, ser. MMSys ’18. Amsterdam, Netherlands: Association for Computing Machinery, 2018.
- [4] ITU-R, “IMT Vision – Framework and overall objectives of the future development of IMT for 2020 and beyond,” International Telecommunication Union, Tech. Rep. Recommendation ITU-R M.2083-0, 2015.
- [5] M. Mozaffari, W. Saad, M. Bennis, Y.-H. Nam, and M. Debbah, “A tutorial on UAVs for wireless networks: Applications, challenges, and open problems,” *IEEE Communications Surveys & Tutorials*, vol. 21, no. 3, pp. 2334–2360, 2019.
- [6] Y. Zeng, Q. Wu, and R. Zhang, “Accessing from the sky: A tutorial on UAV communications for 5G and beyond,” *Proceedings of the IEEE*, vol. 107, no. 12, pp. 2327–2375, 2019.
- [7] T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2002.
- [8] 3GPP, “Study on channel model for frequencies from 0.5 to 100 GHz,” 3rd Generation Partnership Project, Tech. Rep. TR 38.901 V19.2.0, 2025. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.901/
- [9] —, “Enhanced LTE support for aerial vehicles,” 3rd Generation Partnership Project, Tech. Rep. TR 36.777 V15.0.0, 2017. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/36_series/36.777/
- [10] free5GC Project, “free5GC: Open Source Core Network Implementation,” <https://free5gc.org/>, 2026, accessed: 2026-06-21.
- [11] Open5GS Project, “Open5GS: Open Source Implementation for 5G Core and EPC,” <https://open5gs.org/>, 2026, accessed: 2026-06-24.

- [12] A. Patel, “Hexagonal grids,” <https://www.redblobgames.com/grids/hexagons/>, 2013, red Blob Games.
- [13] 3GPP, “Evolved Universal Terrestrial Radio Access (E-UTRA); Base Station (BS) radio transmission and reception,” 3rd Generation Partnership Project, Tech. Rep. TS 36.104, 2025. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/36_series/36.104/
- [14] —, “Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) radio transmission and reception,” 3rd Generation Partnership Project, Tech. Rep. TS 36.101, 2025. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/36_series/36.101/
- [15] —, “Evolved Universal Terrestrial Radio Access (E-UTRA); Physical channels and modulation,” 3rd Generation Partnership Project, Tech. Rep. TS 36.211, 2026. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/36_series/36.211/
- [16] —, “Evolved Universal Terrestrial Radio Access (E-UTRA); Physical layer; Measurements,” 3rd Generation Partnership Project, Tech. Rep. TS 36.214, 2025. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/36_series/36.214/
- [17] —, “NR; Radio Resource Control (RRC) protocol specification,” 3rd Generation Partnership Project, Tech. Rep. TS 38.331, 2026. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.331/
- [18] F. Bai, N. Sadagopan, and A. Helmy, “The IMPORTANT framework for analyzing the impact of mobility on performance of routing protocols for ad hoc networks,” *Ad Hoc Networks*, vol. 1, no. 4, pp. 383–403, 2003.
- [19] 3GPP, “NG-RAN; NG Application Protocol (NGAP),” 3rd Generation Partnership Project, Tech. Rep. TS 38.413, 2026. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.413/